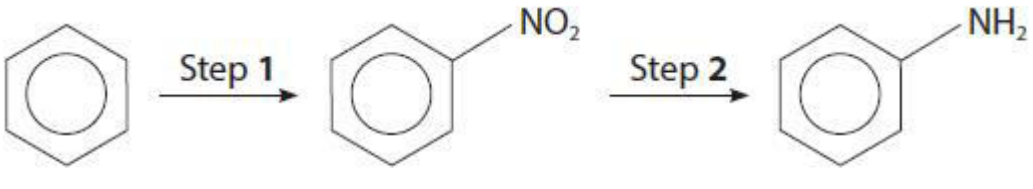
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Questions

Q1.

This question is about aromatic amines and amino acids.

(a) A synthesis of phenylamine is shown.

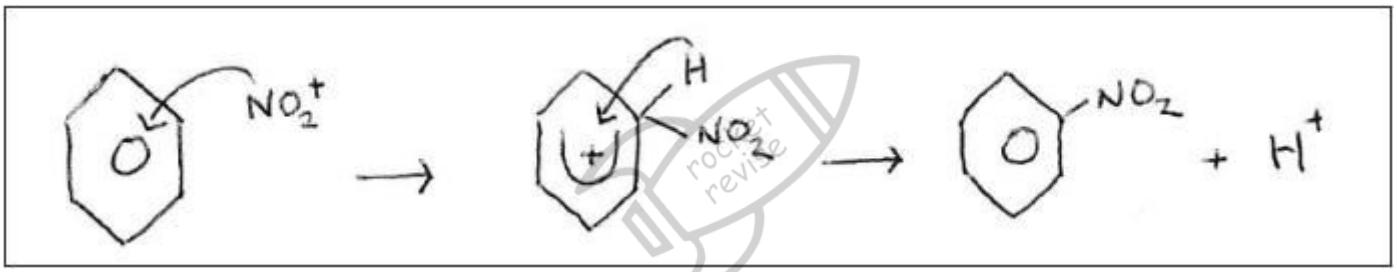


nitrobenzene phenylamine

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(i) Benzene is converted into nitrobenzene in Step 1 using a mixture of concentrated nitric acid and concentrated sulfuric acid. The electrophile is NO_2^+.

A student drew a mechanism for the reaction.



Describe the three changes needed to correct this student's mechanism.

(3)

Change 1 .....

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Change 2 .....

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Change 3 .....

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(ii) Give the reagents for Step **2**.

(1)

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(b) Phenylamine is a base.

(i) Explain why phenylamine is a weaker base than ammonia.

(3)

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(ii) State what is **seen** when a few drops of an aqueous solution of phenylamine are added to an aqueous solution containing Cu^{2+} ions.

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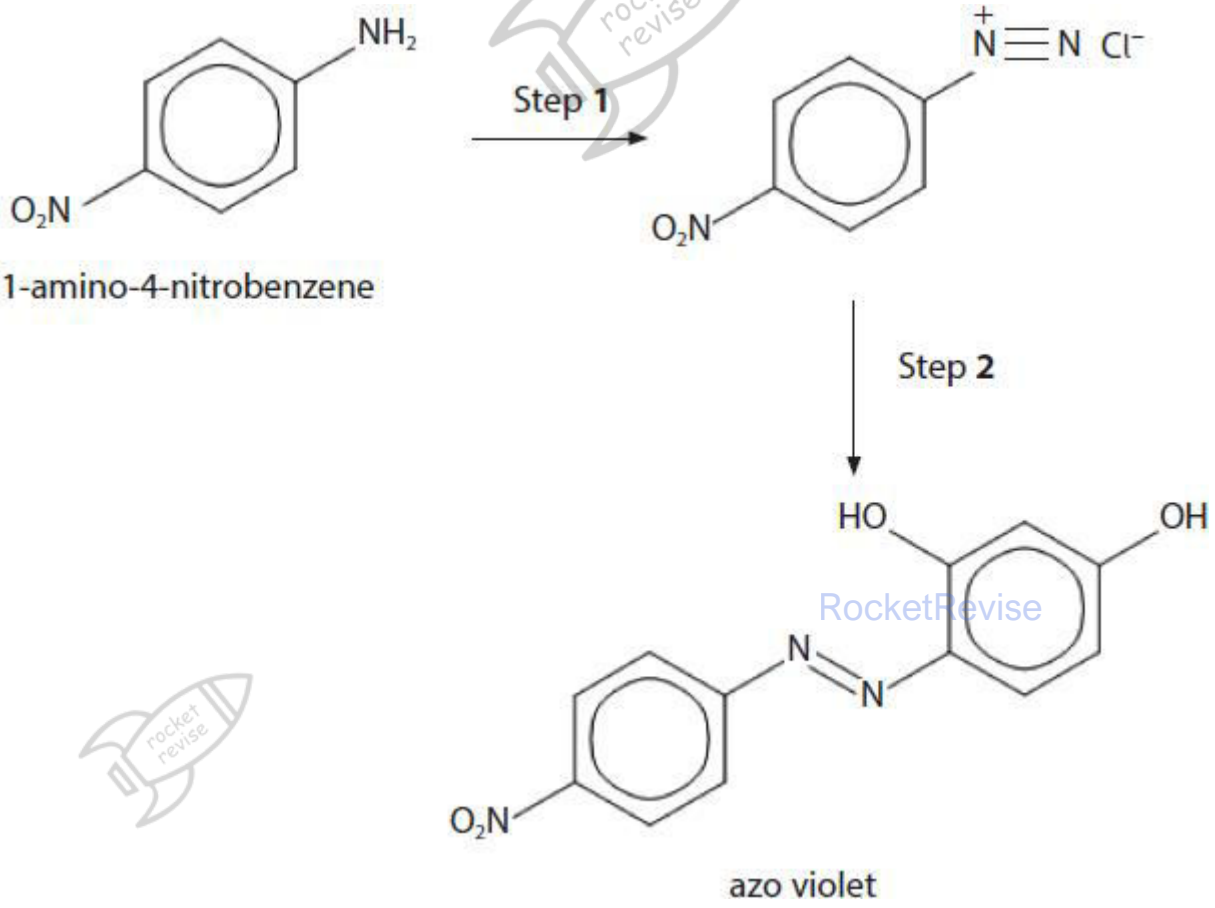
(1)

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(c) Azo violet is used as a dye and an indicator.

A synthesis of azo violet from 1-amino-4-nitrobenzene is shown.



1-amino-4-nitrobenzene

azo violet

(i) Give the reagents and condition for Step **1**.

(2)

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(ii) Deduce the structure of the organic compound needed to produce azo violet in Step **2**.

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(1)

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(iii) Give a reason why azo violet exists as geometric isomers.

(1)

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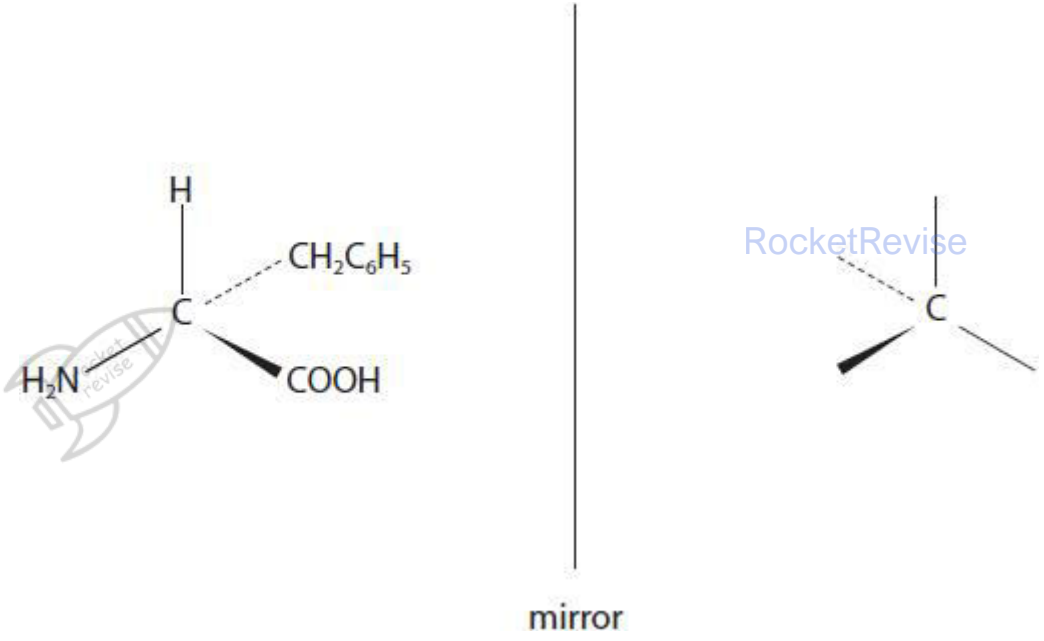
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(d) Phenylalanine is an amino acid that exists as optical isomers.

The structure of one of the optical isomers is shown.
Complete the diagram to show the other optical isomer of phenylalanine.

(1)



(Total for question = 13 marks)

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Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A solution contains 19.6 g dm^{-3} of chromium(III) sulfate, $Cr_2(SO_4)_3$.

[Molar mass $Cr_2(SO_4)_3$ = 392 g mol^{-1}]

- ☐ **A** 0.05
- ☐ **B** 0.10
- ☐ **C** 0.15
- ☐ **D** 0.25

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(Total for question = 1 mark)

Q3.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The electronic configuration of the atoms of an element is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$.

What is the maximum oxidation number shown in compounds of this element?

- ☐ **A** +1
- ☐ **B** +5
- ☐ **C** +6
- ☐ **D** +7

(Total for question = 1 mark)

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Q4.

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The Lanthanides

The lanthanides form a series in the Periodic Table comprising the elements from lanthanum, La ($Z = 57$) to lutetium, Lu ($Z = 71$). They have important uses as elements and compounds, for example, in the screens, circuits and speakers of smartphones. These elements have some similar properties to d-block elements.

Many of the lanthanide elements are obtained by heating their trichlorides or trifluorides with calcium at 1000°C.

These elements form part of the f-block of the Periodic Table as the last electron added enters an f-orbital. There are seven f-orbitals in the f-subshell.

The electronic configurations of the atoms consist of the electronic configuration of xenon followed by electrons in the 4f and 6s orbitals and sometimes the 5d orbitals. For example:

samarium, Sm	$[Xe]4f^6 6s^2$
europium, Eu	$[Xe]4f^7 6s^2$
gadolinium, Gd	$[Xe]4f^7 5d^1 6s^2$
terbium, Tb	$[Xe]4f^9 6s^2$

All of the lanthanides form ions with a charge of 3+ and the ionic radii of these ions decrease as the atomic number of the element increases.

The lanthanides form complex ions, for example, $[Ce(H_2O)_9]^{3+}$ and $[Yb(H_2O)_8]^{3+}$.

Many of the complex ions formed from the elements in the lanthanide series are coloured and this colour is caused by f–f transitions.

Cerium(IV) ammonium nitrate is yellow in aqueous solution and gives a red colour with **all** alcohols.

Cerium(IV) ammonium sulfate is used in redox titrations with ferroin as the indicator. The colour change is red to yellow.

Use your knowledge of d-block elements to help you to answer the questions.

(a) Write the equation for the reduction of holmium(III) fluoride, HoF_3, by calcium.

State symbols are not required.

(1)

(b) The electronic configuration of xenon is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^6$.

(i) Suggest a reason why gadolinium has the electronic configuration $[Xe]4f^7 5d^1 6s^2$ rather than $[Xe]4f^8 6s^2$.

(1)

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(ii) Complete the electronic configuration of Sm^{3+}.

(1)

[Xe] .....

(c) (i) Explain why the ionic radius of a thulium ion, Tm3+, is less than that of a cerium ion, Ce^{3+}.

Tm^{3+} 0.095 nm

Ce^{3+} 0.111 nm

(2)

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(ii) Suggest a reason why the complex ions of the lanthanides can contain more ligands than those of the transition metals.

(1)

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(d) Explain why solutions containing $[La(H_2O)_9]^{3+}$

(2)

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(e) Hydrated cerium(IV) ammonium nitrate contains Ce^{4+}, NH_4^+ and NO_3^- ions and some water of crystallisation.

(i) Hydrated cerium(IV) ammonium nitrate contains 23.97 % cerium, 19.18 % nitrogen, 2.05 % hydrogen
and 54.80 % oxygen by mass.
Calculate the empirical formula of this compound and hence write the overall formula showing the cerium,
ammonium and nitrate ions and the water of crystallisation.

(3)

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(ii) An organic compound, **X**, with the molecular formula $C_7H_{16}O$ did not react with acidified potassium dichromate(VI) solution but gave a red colour with cerium(IV) ammonium nitrate solution. **X** exists as a pair of optical isomers.

Deduce a possible structure of **X**. Justify your answer.

(4)

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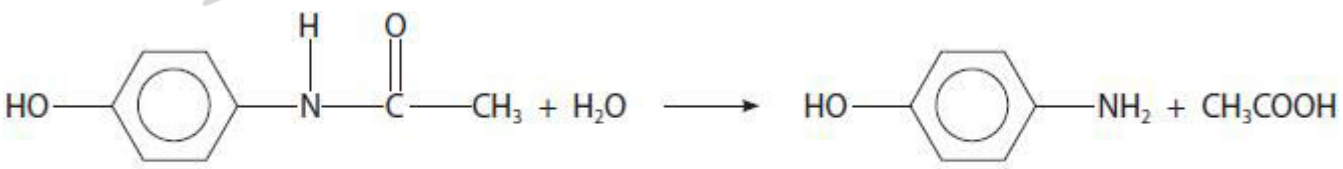
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(f) The amount of paracetamol in a tablet can be determined using a titration with cerium(IV) ions.

The tablets are crushed and then hydrolysed in acid to form 4-aminophenol.

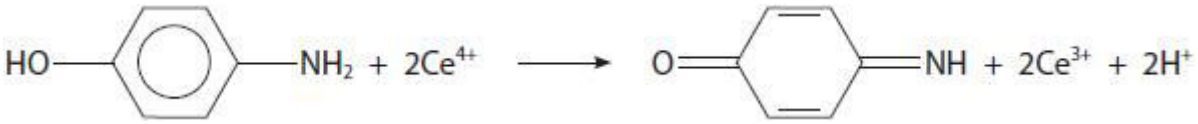


paracetamol 4-aminophenol

4-aminophenol is oxidised by cerium(IV) ions.

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Outline procedure

- Two tablets containing paracetamol were crushed and 0.800 g of the powder was added to dilute sulfuric acid.
- The mixture was heated under reflux until the hydrolysis was complete.
- The solution was made up to 100.0 cm^3 in a volumetric flask.
- 25.0 cm^3 portions of the solution were titrated against acidified 0.100 mol dm^{-3} Ce^{4+} using ferroin the indicator.

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Result

The mean titre was 21.70 cm^3.

Calculate the percentage by mass of paracetamol in the tablets.

Give your answer to an appropriate number of significant figures.

(5)

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(Total for question = 20 marks)

Q5.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of the following species will **not** act as a ligand in the formation of a complex ion?

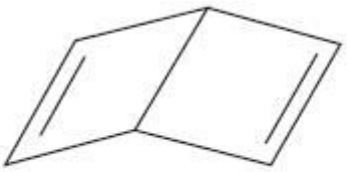
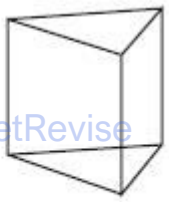
- ☐ **A** CH_3NH_2
- ☐ **B** CN^-
- ☐ **C** NH_3
- ☐ **D** NH_4^+

(Total for question = 1 mark)

Q6.

This question is about benzene, C_6H_6, a colourless liquid first isolated in 1825, and some related compounds.

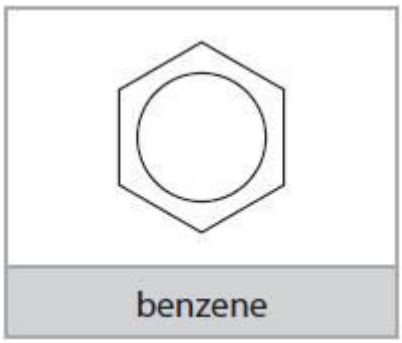
Three C_6H_6 structures proposed in the 1860s are shown.

		
Kekulé structure	Dewar structure	Ladenburg structure

The delocalised model for the structure of benzene has been accepted since the 1930s following the study of its X-ray diffraction pattern and the understanding of electron delocalisation in bonding theory.

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benzene

The Dewar and Ladenburg structures have since been isolated as stable compounds but there is no compound with the Kekulé structure.

(a) Describe a **chemical** test, including the result, that could distinguish the Dewar structure from benzene.

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(2)

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(b) State **one** similarity and **one** difference you would expect in the **low** resolution proton NMR spectrum of the Ladenburg structure and that of benzene.

You **must** include data from the Data Booklet to support your answer.

(2)

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(c) Explain how X-ray diffraction shows that benzene has a delocalised structure and **not** a Kekulé structure.

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(2)

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(d) The Ladenburg and Dewar structures both isomerise to benzene.

The enthalpy changes are –376 kJ mol^{-1} and –297 kJ mol^{-1} respectively.

(i) Draw a **labelled** enthalpy level diagram showing the relative thermodynamic stability of the

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Ladenburg structure, the Dewar structure and benzene.
Include the enthalpy change values in kJ mol^{-1}.
Your diagram does **not** need to be to scale.

(2)



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(ii) Give a possible reason why the isomerisation of the Dewar structure to benzene has a lower activation energy than that of the Ladenburg structure to benzene.

(1)

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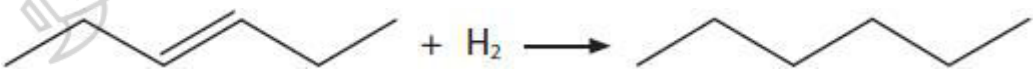
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(e) The enthalpy change of hydrogenation of hex-3-ene is –118 kJ mol^{-1}.



The table shows the enthalpy changes of hydrogenation of two further alkenes containing six carbon atoms.

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Alkene	Enthalpy change of hydrogenation / kJ mol^{-1}
 *E*-hexa-1,4-diene	−236
 *E*-hexa-1,3-diene	−214

Use your knowledge of benzene thermochemistry to suggest explanations for **both** of these enthalpy changes of hydrogenation in relation to the value for hex-3-ene.

(2)

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(f) Methylbenzene, $C_6H_5CH_3$, reacts with ethanoyl chloride, CH_3COCl, in the presence of the catalyst aluminium chloride, $AlCl_3$, to form a mixture of organic products with the formula $CH_3COC_6H_4CH_3$.

$$C_6H_5CH_3 + CH_3COCl \rightarrow CH_3COC_6H_4CH_3 + HCl$$

(i) Draw the **skeletal** formulae of **three** different arenes with the formula $CH_3COC_6H_4CH_3$.

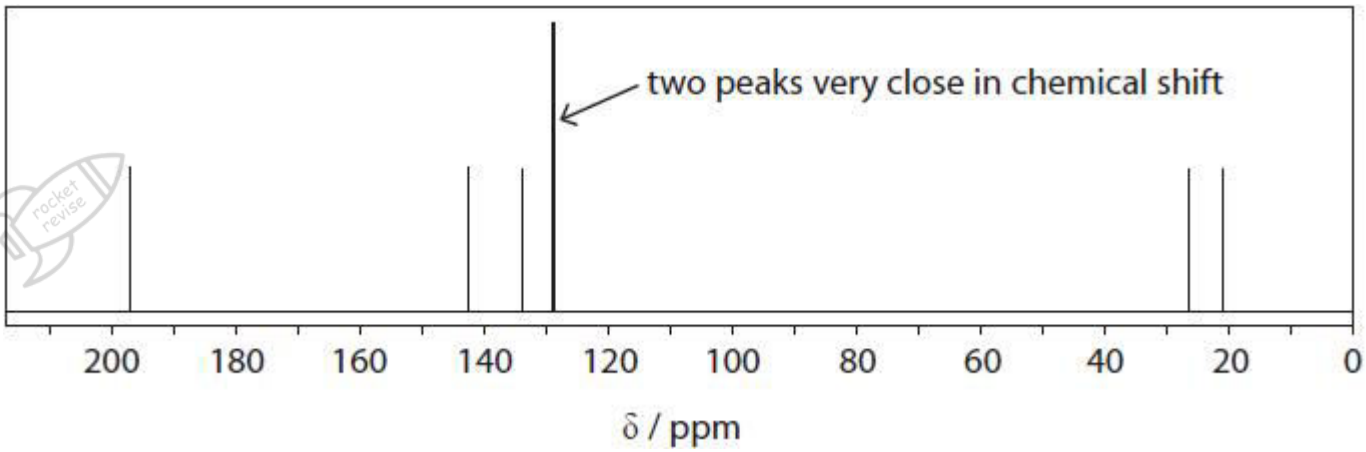
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(ii) The ^{13}C NMR spectrum of **one** of these arenes, compound **X**, is shown.

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Identify compound **X**.
Use the number of peaks on the ^{13}C NMR spectrum to justify your answer.

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(2)

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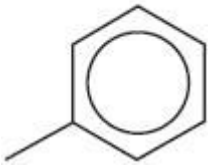
(iii) Complete the diagram, including curly arrows, to show the mechanism for the formation of compound **X** in this reaction.
Include an equation for the regeneration of the catalyst.

(4)

$$CH_3COCl + AlCl_3 \rightarrow CH_3CO^+ + AlCl_4^-$$

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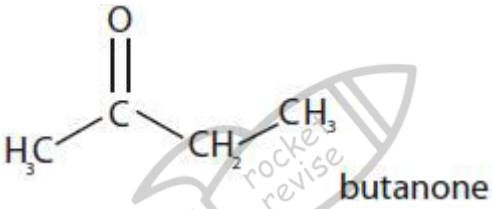
(Total for question = 19 marks)

Q7.

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Ketones are useful starting compounds in organic synthesis.

This question is about butanone.



butanone

(a) The mass spectrum of butanone has significant peaks at $m / z = 43$ and at $m / z = 57$.

(i) Give the structures of the species responsible for these two peaks.

(2)

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(ii) Give the structure of **one** other species that you would expect to produce a peak at a different m / z value in the mass spectrum of butanone.

(1)

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(b) Devise a reaction scheme to prepare propan-1-ol from butanone, using no more than **four** steps.

Identify the reagents and essential conditions for each step and give the name or structure of each of the intermediate compounds.

(4)

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(c) Devise a reaction scheme to prepare 2-methylbut-2-ene from butanone, using no more than **four** steps.

Identify the reagents and essential conditions for each step and give the name or structure of each of the intermediate compounds.

(5)

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(Total for question = 12 marks)

Q8.

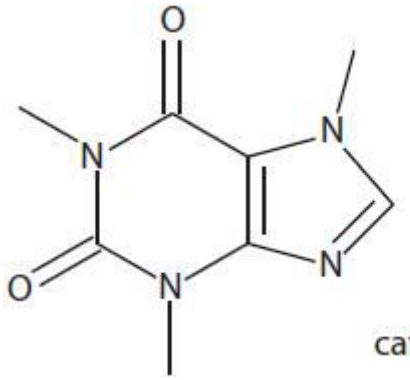
Coffee Chemistry

There are over a thousand chemical compounds in coffee and their physiological effects are the subject of considerable speculation and research. The verdict on coffee is contradictory: some of the compounds have been identified as toxic and even carcinogenic but others are antioxidants associated with cancer prevention. Recent research has identified compounds in coffee that might be used in the treatment of prostate cancer.

By far the best known compound in coffee is caffeine, the most widely consumed psychoactive drug in the world. In small amounts it is a stimulant but doses in excess of 10 g per day are toxic. Caffeine contains amide and amine groups.

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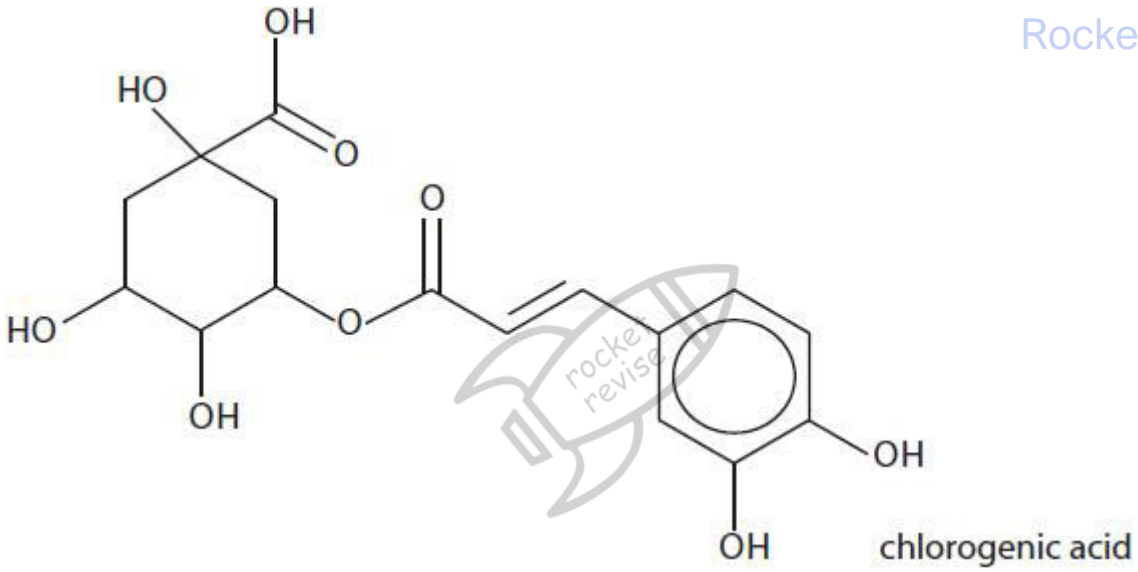
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caffeine

Chlorogenic acid is responsible for the acidic taste of coffee. It is an antioxidant and has also been shown to slightly decrease blood pressure.

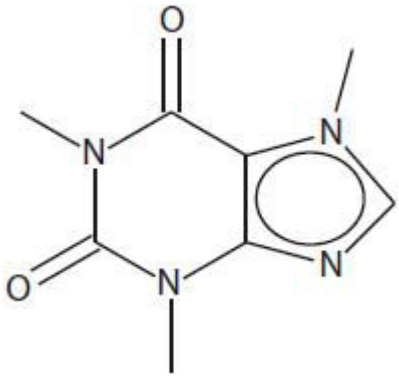
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chlorogenic acid

Caffeic acid, quinic acid and acetoin are also present in coffee.

(a) Another way of drawing the structure of caffeine is shown.



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(i) The bonding represented by this diagram of caffeine differs from that given in the passage. Explain what this diagram indicates about the bonding in caffeine, stating the effect on the structure of caffeine.

(3)

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(ii) Suggest why caffeine is a much weaker base than a primary amine such as ethylamine, even though the right-hand ring has two amine groups.

(2)

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(b) A 200 cm^3 cup of coffee contains approximately 85 mg of caffeine.

(i) Calculate the concentration, in mol dm^{-3}, of caffeine in this cup of coffee.
Give your answer to an appropriate number of significant figures.

(4)

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(ii) The removal of caffeine from the body is a first order reaction with a half-life of between three and seven hours for an adult.

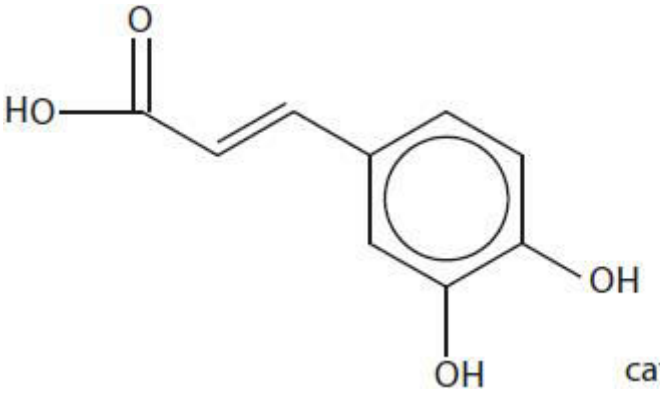
An adult drinks coffee containing a total of 160 mg of caffeine.
Calculate to the nearest hour the **minimum** time needed for the amount of caffeine in their body to drop to 20 mg.

(2)

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(c) Chlorogenic acid is an ester of caffeic acid, a compound that is present in all plants.



caffeic acid

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(i) A student suggested that caffeic acid could be synthesised by an electrophilic substitution of 1,2-dihydroxybenzene.

Draw the mechanism of this electrophilic substitution, including the formation of a suitable electrophile.

(5)

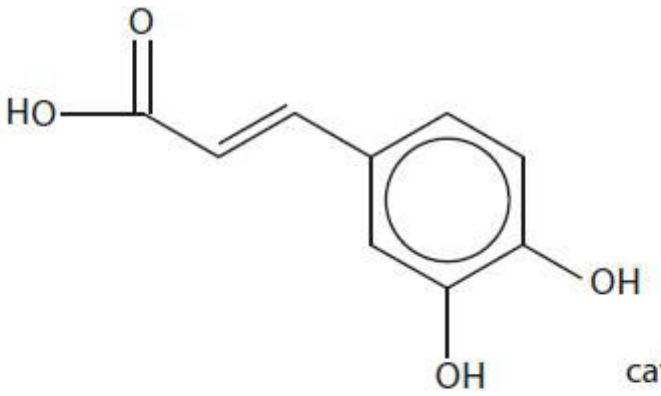
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(ii) Deduce the structure of quinic acid which combines with caffeic acid to form chlorogenic acid.

(1)

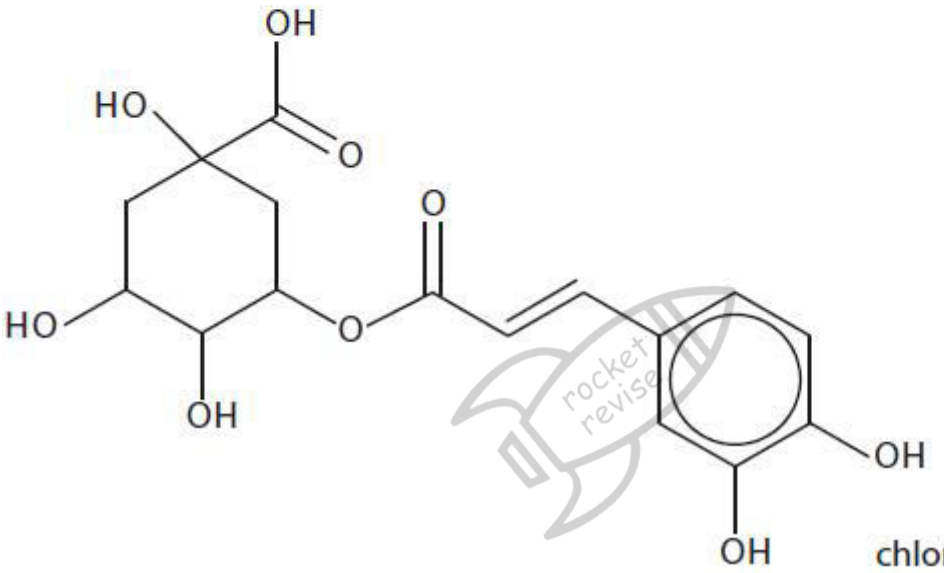
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caffeic acid

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chlorogenic acid

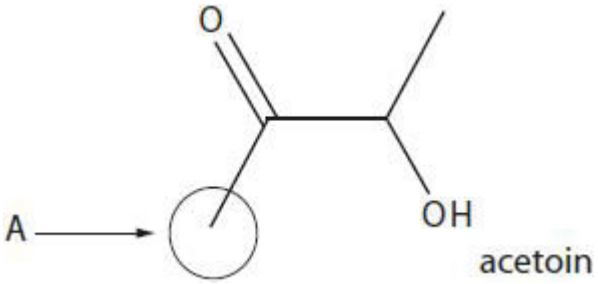
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quinic acid

(d) The structure of acetoin is shown with one of the proton environments labelled.

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acetoin

(i) Identify the other proton environments of acetoin on the structure and label them B, C etc.

(1)

(ii) Complete the table to show the splitting pattern in the high resolution proton NMR spectrum of acetoin.

(2)

Proton environment	Splitting pattern
A	

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(Total for question = 20 marks)

Q9.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The melting temperature is determined for impure crystals of an organic compound. When compared with a data book value for the pure compound, the measured melting temperature

- ☒ **A** will be the same as the true value
- ☒ **B** will be higher than the true value
- ☒ **C** will be lower than the true value
- ☒ **D** could be higher or lower than the true value

(Total for question = 1 mark)

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Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Grignard reagents react with

- ☒ **A** water giving primary alcohols
- ☒ **B** all aldehydes giving secondary alcohols
- ☒ **C** ketones giving secondary or tertiary alcohols
- ☒ **D** carbon dioxide giving carboxylic acids

(Total for question = 1 mark)

Q11.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

In the **high** resolution proton NMR spectrum of propan-2-ol, $CH_3CHOHCH_3$ there are

- ☒ **A** one singlet, one doublet and a heptet
- ☒ **B** one singlet, two doublets and a heptet
- ☒ **C** two singlets and two triplets
- ☒ **D** three singlets and a quartet

(Total for question = 1 mark)

Q12.

Using excess oxygen, 25 cm^3 of a gaseous hydrocarbon C_xH_y was burned completely.

After cooling to room temperature the total gas volume was measured and found to be 75 cm^3 less than the total gas volume before the mixture was ignited.

When the product gases were shaken with potassium hydroxide solution, the total gas volume decreased by a further 100 cm^3.

All gas volumes were measured at room temperature and pressure.

A general equation for the combustion of a hydrocarbon is

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$$C_xH_y + \left(x + \frac{y}{4}\right) O_2 \rightarrow xCO_2 + \left(\frac{y}{2}\right) H_2O$$

(a) Determine the molecular formula of C_xH_y. You **must** show your working.

(3)

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(b) When C_xH_y was added to a little bromine water and the mixture shaken, the bromine water remained yellow.

Suggest **two** possible structures for C_xH_y .

(2)

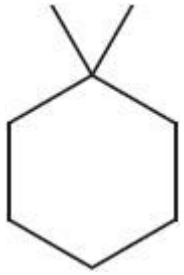
(Total for question = 5 marks)

Q13.

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Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The structure of a hydrocarbon is shown.



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How many peaks will there be in the ^{13}C NMR spectrum of this compound?

☒ **A** four

☒ **B** five

☒ **C** seven

☒ **D** eight

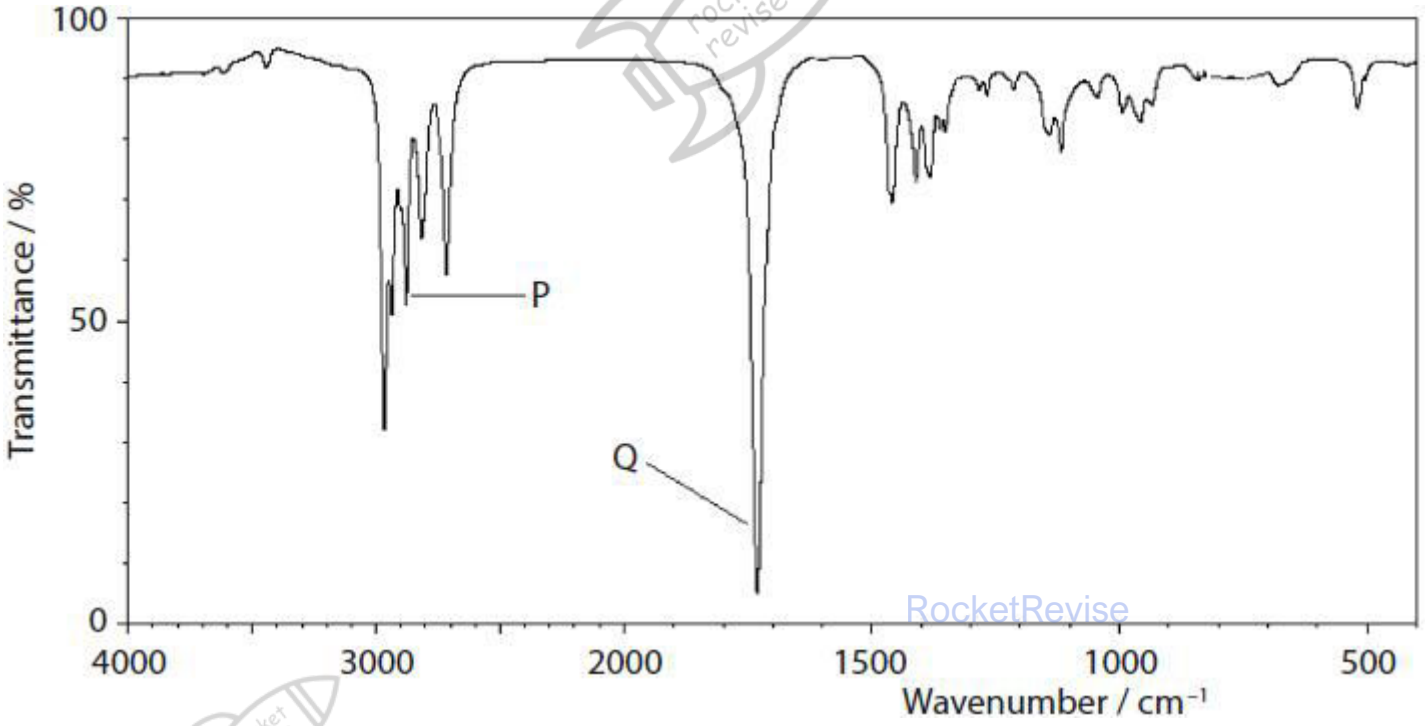
(Total for question = 1 mark)

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Q14.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

An aliphatic organic compound has the infrared spectrum shown.



What are the bond stretches responsible for the peaks **P** and **Q** in the spectrum?

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	P	Q
☐ A	O—H carboxylic acid	C═O carboxylic acid
☐ B	O—H carboxylic acid	C═O aldehyde
☐ C	C—H aldehyde	C═O carboxylic acid
☐ D	C—H aldehyde	C═O aldehyde

(Total for question = 1 mark)

RocketRevise

Q15.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

When a sample of a hydrocarbon is burned completely in oxygen, 2.64 g of carbon dioxide and 0.81 g of water are formed.

Which of these could be the **molecular** formula of the hydrocarbon?

☐ **A** C_2H_3

☐ **B** C_4H_3

☐ **C** C_4H_6

☐ **D** $C_{12}H_9$

(Total for question = 1 mark)

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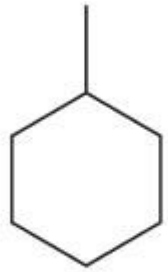
Q16.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

How many peaks are there in the carbon-13 (^{13}C) NMR spectrum of methylcyclohexane?

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methylcyclohexane

- ☐ **A** one
- ☐ **B** three
- ☐ **C** five
- ☐ **D** seven

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(Total for question = 1 mark)

Q17.

A series of reactions with iron and iron complexes was carried out.

Reaction **1**

A sample of iron was heated with hydrochloric acid and a pale green hydrated salt **A** with molar mass 198.8 g mol^{-1} was crystallised from the solution.

Reaction **2**

Salt **A** was dissolved in water forming a pale green solution containing complex ion **B**. On addition of excess aqueous potassium cyanide, KCN, the solution turned yellow due to the formation of complex ion **C**.

Reaction **3**

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Chlorine gas was bubbled through the solution containing complex ion **C** forming a red solution of complex ion **D**.
Salt **E**, the potassium salt of complex ion **D**, was then crystallised from the solution.

(a) Deduce the formula of the hydrated salt **A**.

You **must** show your working.

(2)

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.....

(b) Give the **formula** of complex ion **B**.

(1)

.....

(c) Complex ion **C** has six cyanide ligands.

Draw the structure of **C**, clearly showing its three-dimensional shape.

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(1)

(d) The percentage composition by mass of salt **E** is

K = 35.6% Fe = 17.0% C = 21.9% N = 25.5%

Calculate the empirical formula of salt **E**.

(3)

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(e) Write the **ionic** equation for the reaction of complex ion **C** with chlorine to form complex ion **D**.

(2)

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(f) Complete the table, using ✓ as appropriate, to identify the type of each reaction.

(2)

	Neutralisation	Ligand exchange	Redox
Reaction 2			
Reaction 3			

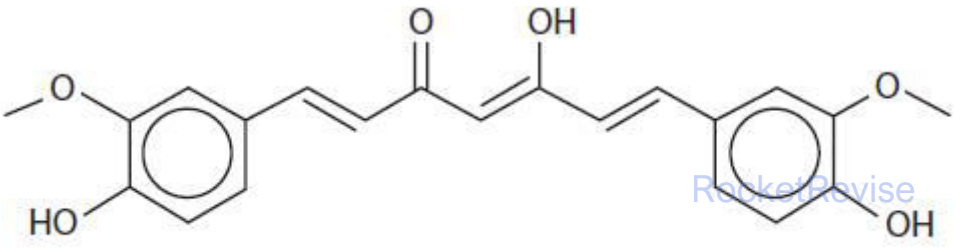
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(Total for question = 11 marks)

Q18.

Organic molecules are an important source of colour both in the natural world and in a wide range of industrial applications.

Curcumin contributes to the yellow colour of turmeric spice and is used as an additive in cosmetics and foods. It has been suggested that curcumin can act as an antioxidant and anticancer agent, through reactions with free radicals and proteins, and may also inhibit Alzheimer's disease by complexing to toxic metal ions.

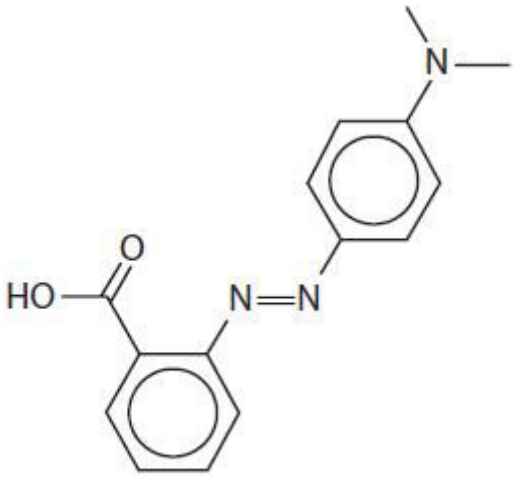


curcumin

Azo dyes are synthetic compounds that do not occur naturally. They can be used to colour textiles such as cotton. The acid-base indicator methyl red is an azo dye.

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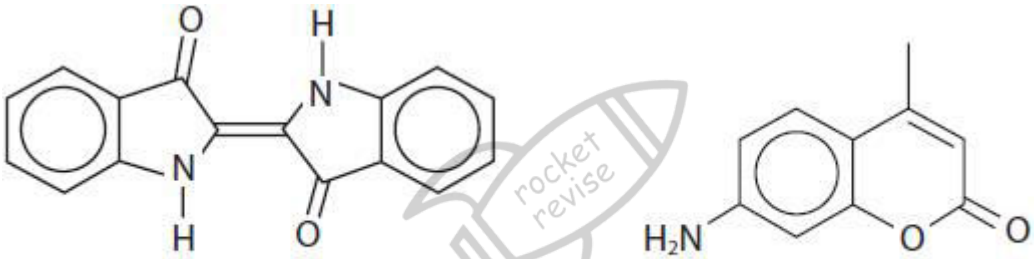
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methyl red

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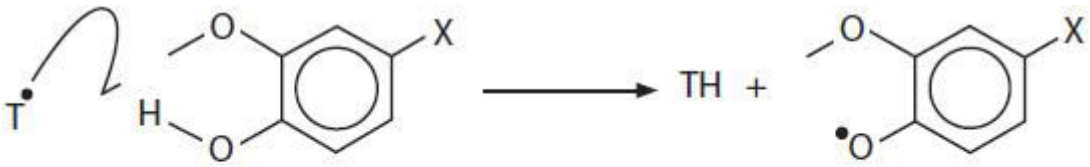
Indigotin is used to dye denim a blue colour and coumarin 440 is used to generate blue light in lasers. Both dyes occur naturally in plants but can be synthesised in the laboratory.



indigotin

coumarin 440

(a) The equation shows the reaction between a free radical T• and curcumin (shown by a simplified structure).



curcumin

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Complete the left-hand side of the equation by adding curly half-arrows.

(1)

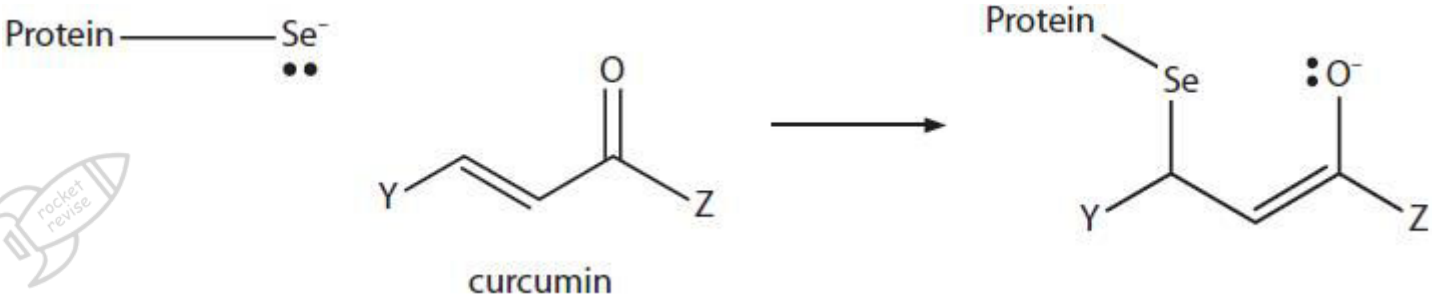
(b) Selenide anions attached to protein side-chains may undergo nucleophilic addition reactions with curcumin.

Complete the mechanism for one of the steps in such a reaction, adding curly arrows to the simplified structures shown.

(2)

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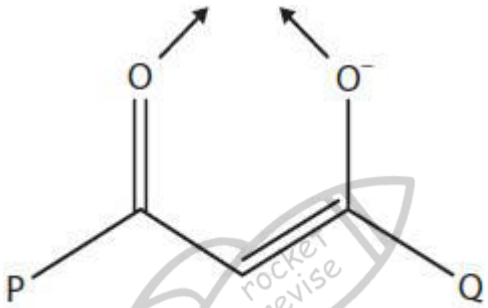
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curcumin

(c) Curcumin anions can act as bidentate ligands in metal-curcumin (M-curc) complexes. The oxygen atoms of the curc ligand occupy adjacent coordination sites in the complex, as shown.

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curcumin anion

Complete the table relating to two M-curc complexes.

(2)

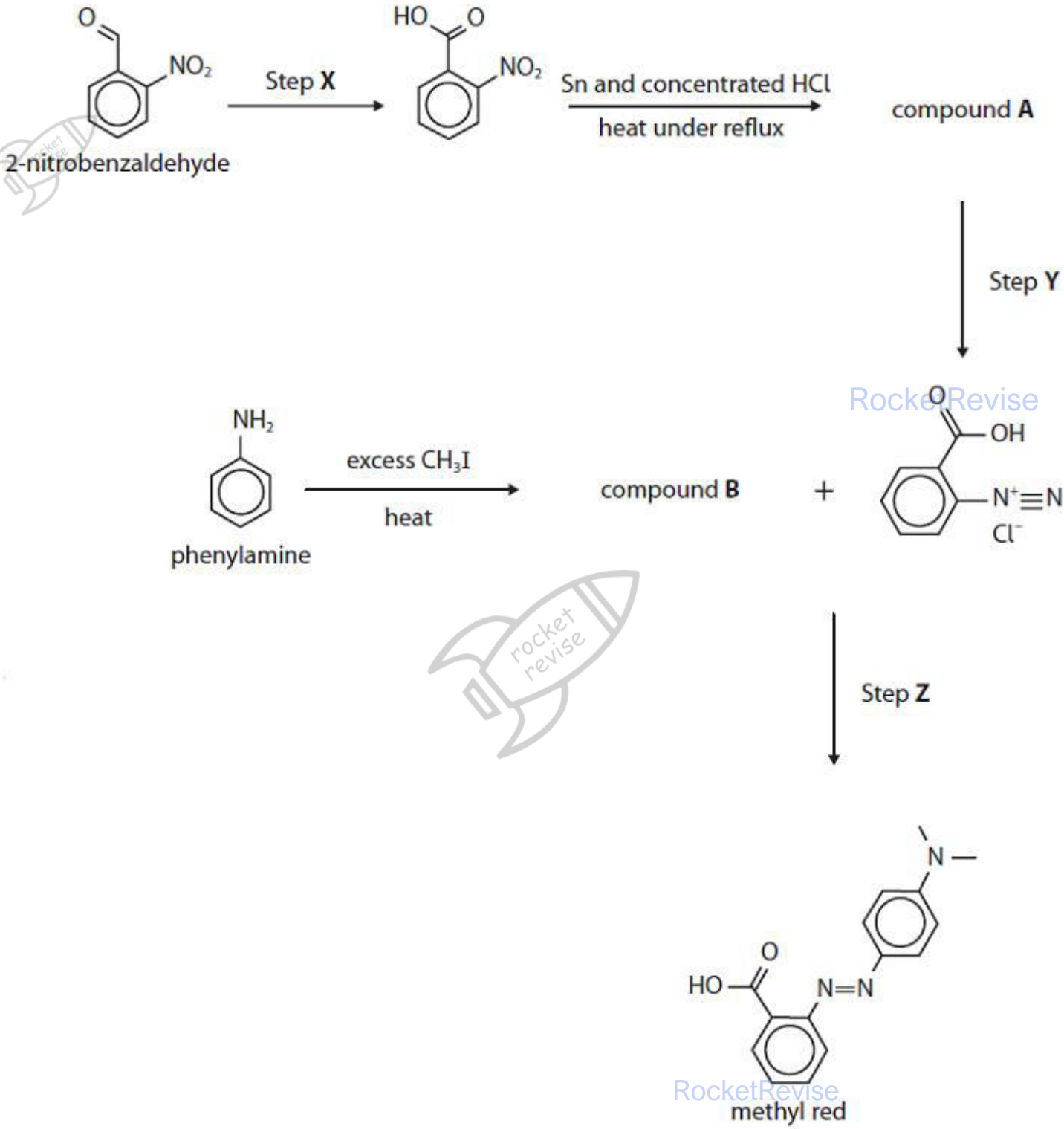
	$[Au(curc)_2]^+$	$[Al(curc)(C_2H_5OH)_2(NO_3)_2]$
Coordination number	4	
O—M—O bond angle	90°	
Shape		octahedral
Charge on metal ion		+3

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(d) An incomplete synthesis for methyl red starting from 2-nitrobenzaldehyde and phenylamine is shown.

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(i) State the reagents and conditions needed in Step **X**.

(2)

.....

.....

(ii) Draw the structure of compound **A**.

(1)

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(iii) State the reagents needed in Step **Y**.

(1)

.....

.....

(iv) Draw the structure of compound **B**.

(1)

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(v) The temperature used in Steps **Y** and **Z** should be kept as close to 5 °C as possible. State why the temperature should be neither higher nor lower than 5 °C.

(2)

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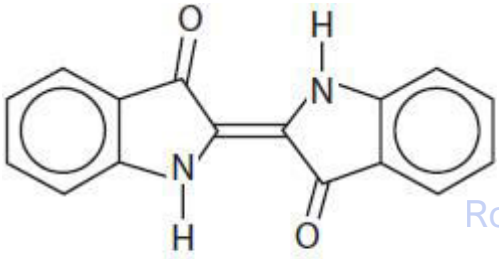
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(e) Indigotin can be synthesised from 2-nitrobenzaldehyde and propanone in aqueous sodium hydroxide.

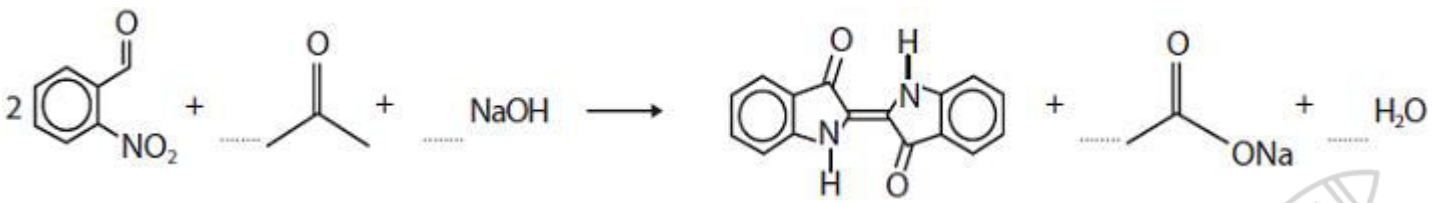


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indigotin

(i) Complete the equation for this reaction.

(2)



(ii) Calculate the mass of 2-nitrobenzaldehyde required to make 10.0 g of indigotin from this reaction when the percentage yield is 85.0 %.

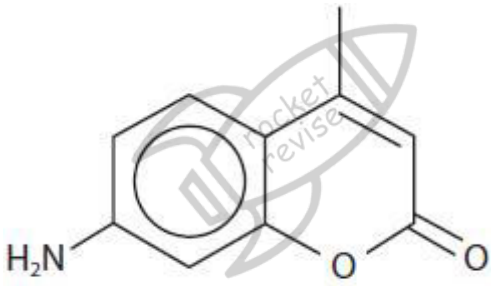
(3)

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(f) Give the structure of the **organic** product of each of the following reactions of coumarin 440.



coumarin 440

(i) Hydrolysis with **excess** aqueous sodium hydroxide.

(2)

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(ii) Condensation with ethanoyl chloride.

(1)

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(Total for question = 20 marks)

Q19.

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Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

What is the minimum volume of oxygen gas, measured at room temperature and pressure, required for the complete combustion of 9.2 g of $C_3H_8O_3$ (M_r = 92)?

[Molar volume of gas at room temperature and pressure = 24.0 dm^3 mol^{-1}]

☐ **A** 4.8 dm^3

☐ **B** 8.4 dm^3

☐ **C** 12.0 dm^3

☐ **D** 16.8 dm^3

(Total for question = 1 mark)

Q20.

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A compound, **Q**, is a pale yellow liquid that is the main constituent of cinnamon oil. **Q** contains the elements carbon, hydrogen and oxygen only.

(a) Complete combustion of 6.02 g of **Q** produces 18.07 g of carbon dioxide and 3.30 g of water.

Determine the empirical formula of **Q**.

(4)

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(b) Tests on samples of **Q** show that it

- burns in air with a very sooty flame
- forms an orange precipitate with Brady's reagent (2,4-dinitrophenylhydrazine solution)
- forms a silver precipitate with Tollens' reagent
- decolourises bromine water
- exists as a pair of geometric isomers.

Deduce a structure for **Q**, explaining how each piece of information supports your answer.

(6)

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(Total for question = 10 marks)

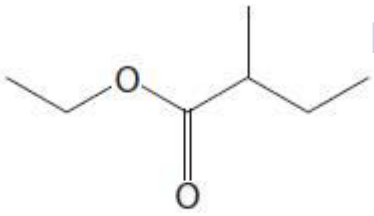
Q21.

The ester ethyl 2-methylbutanoate is found in wild berries such as bilberries.

Devise a synthesis to convert but-2-ene into ethyl 2-methylbutanoate in **four** steps.



but-2-ene



ethyl 2-methylbutanoate

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Include the reagents and essential conditions for each step and the name or structure of each of the intermediate compounds.

Details of practical procedures are not required.

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(Total for question = 6 marks)

Q22.

A compound **Q** contains the elements carbon, hydrogen and oxygen only.

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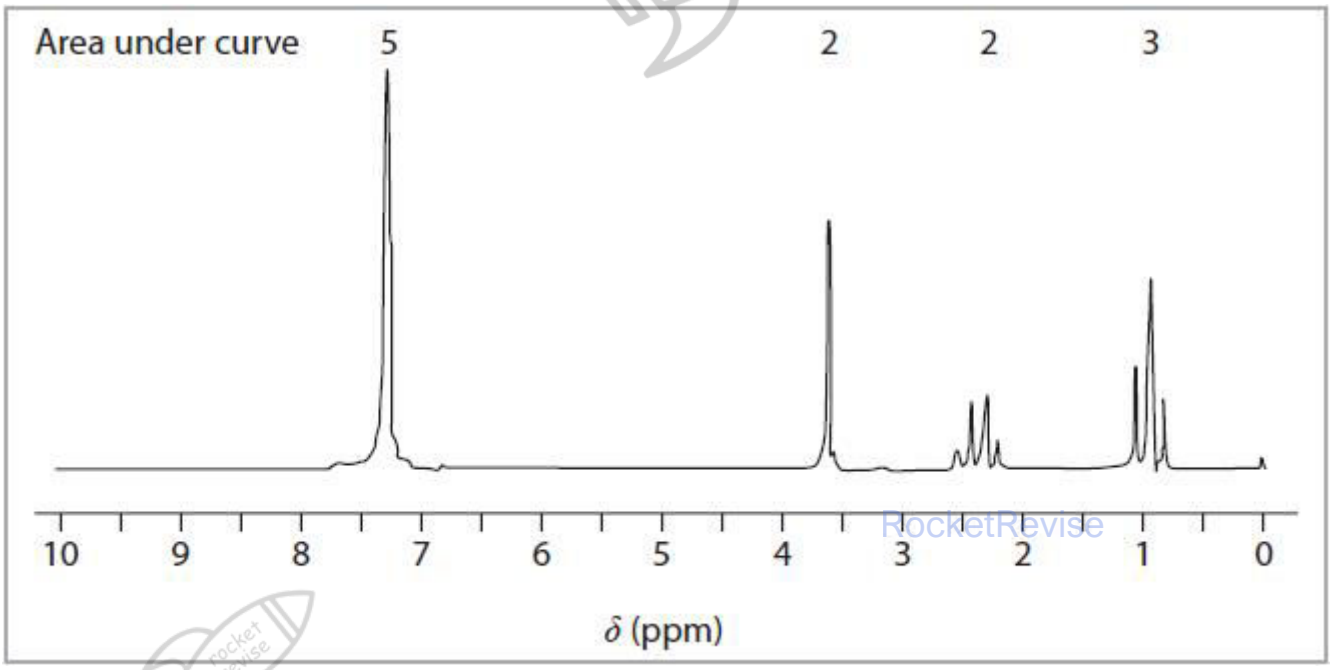
(a) Combustion analysis of 4.91 g of **Q** produces 14.6 g of carbon dioxide and 3.58 g of water.

Show that the molecular formula of **Q** is $C_{10}H_{12}O$.
You **must** show all your working.
[M_r of **Q** = 148]

(4)

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(b) The high resolution proton NMR spectrum of **Q** is shown.



Deduce the structure of **Q**. Justify your answer by considering the relative peak areas, the chemical shifts and the splitting patterns.
You will find it helpful to refer to page 8 of the Data Booklet.
The peak at 3.6 ppm is due to a proton environment on a carbon bonded to the benzene ring. The peak is not where it might be expected from the general values in the Data Booklet.

(7)

(Total for question = 11 marks)

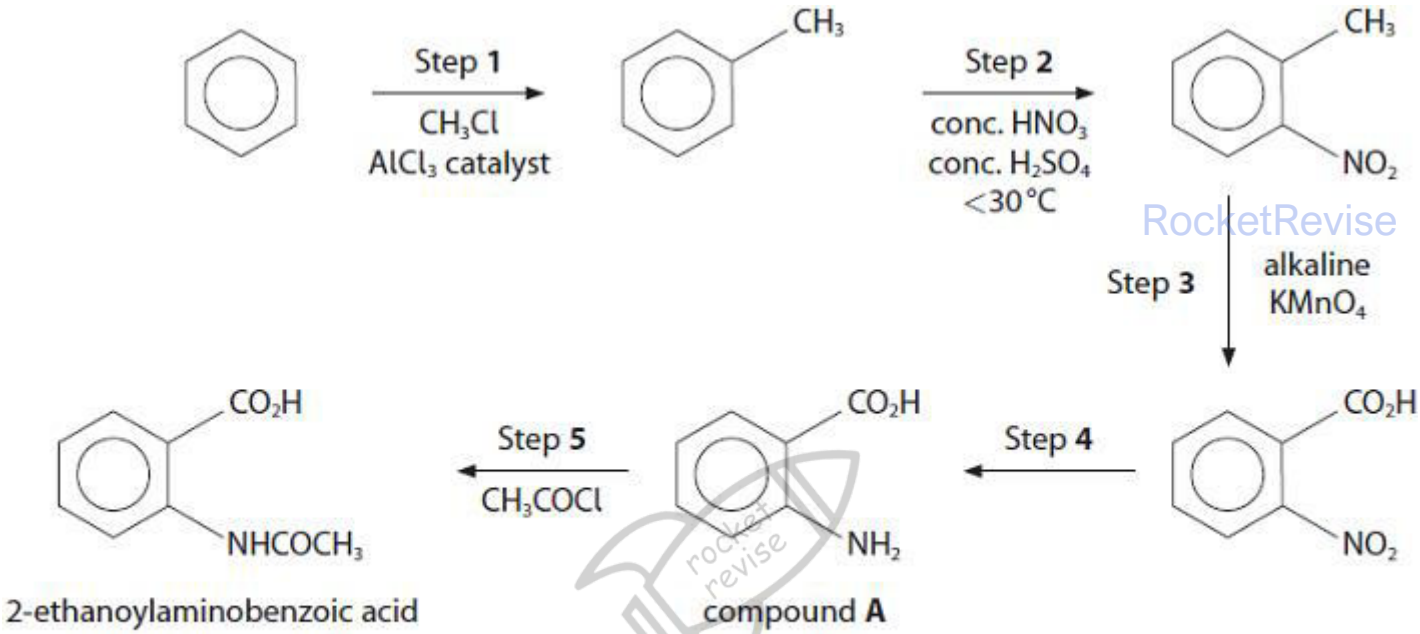
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Q23.

This question is about 2-ethanoylaminobenzoic acid.
Crystals of this compound emit flashes of light when crushed.

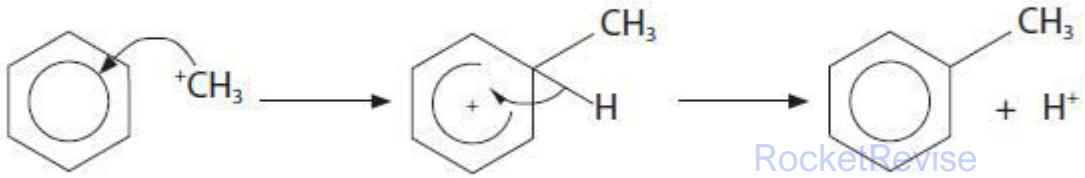
2-ethanoylaminobenzoic acid can be synthesised using benzene as a starting material.



(a) A student proposed a mechanism for Step 1.

(a) A student proposed a mechanism for Step 1.

$$CH_3Cl + AlCl_3 \longrightarrow {}^{+}CH_3 + AlCl_4^-$$



$$H^+ + AlCl_4^- \longrightarrow HCl + AlCl_3$$

(i) Identify **two** errors in the mechanism by circling them.

(2)

(ii) State how the errors identified in (a)(i) should be corrected. Justify your answer in terms of the structures involved.

(2)

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(b) Give a reason why it is important to keep the temperature below 30 °C in Step **2**.

(1)

.....

.....

.....

(c) State the type of reaction in Step **3**.

(1)

.....

(d) Identify, by name or formula, the reagents needed for Step **4**.

(1)

.....

.....

(e) Explain how the structures of ethanoyl chloride, CH_3COCl, and compound **A** enable them to react forming 2-ethanoylaminobenzoic acid in Step **5**.

(3)

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(f) Calculate the volume of benzene required to form 5.92 g of 2-ethanoylaminobenzoic acid, assuming the overall yield for the synthesis is 28.2 %.

Give your answer to an appropriate number of significant figures.
[Density of benzene = 0.879 g cm^{-3}]

(4)

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(Total for question = 14 marks)

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Q24.

Answers the question with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about a Grignard reagent, 2-methylpropylmagnesium bromide.



This Grignard reagent can be prepared by refluxing 1-bromo-2-methylpropane with magnesium in a flask containing a dry solvent and anti-bumping granules.

(a) Which solvent is used in this preparation?

(1)

- ☒ **A** ethanol
- ☒ **B** ether
- ☒ **C** pentane
- ☒ **D** propanone

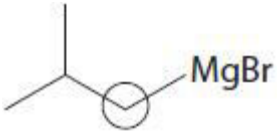
(b) Why are anti-bumping granules added to the flask?

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(1)

- ☒ **A** to lower the boiling temperature of the solvent
- ☒ **B** to prevent the solvent evaporating
- ☒ **C** because the solvent is highly flammable
- ☒ **D** to ensure the solvent boils smoothly

(c) The structure of 2-methylpropylmagnesium bromide is shown.



Which statement best describes the circled carbon atom?

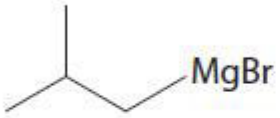
(1)

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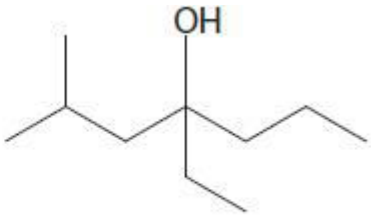
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- ☒ **A** positive and electrophilic
- ☒ **B** positive and nucleophilic
- ☒ **C** negative and electrophilic
- ☒ **D** negative and nucleophilic

(d) Which compound reacts with 2-methylpropylmagnesium bromide to form 4-ethyl-2-methylheptan-4-ol?



2-methylpropylmagnesium bromide



4-ethyl-2-methylheptan-4-ol

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(1)

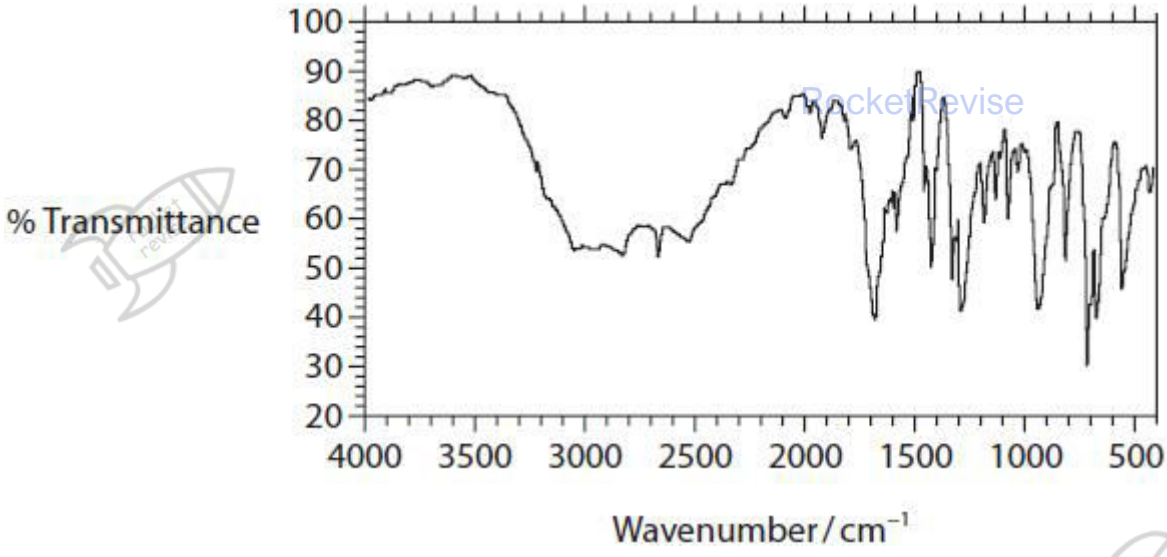
- ☒ **A** hexan-3-one
- ☒ **B** hexan-2-one
- ☒ **C** hexan-3-ol
- ☒ **D** hexan-2-ol

(Total for question = 4 marks)

Q25.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The infrared spectrum of a compound **X** is shown.

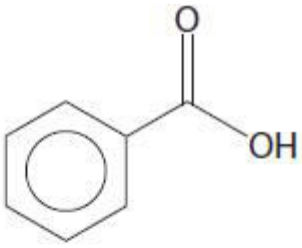


Which could be compound **X**?

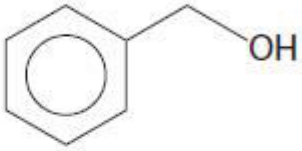
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☐ **A**

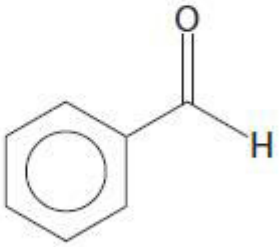


☐ **B**

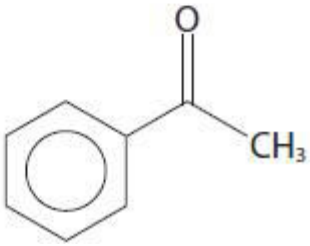


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☐ **C**



☐ **D**



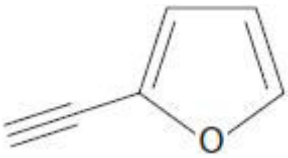
(Total for question = 1 mark)

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Q26.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The mass spectrum of the compound shown is obtained using a high resolution mass spectrometer.



What is the mass to charge ratio, *m* / *z*, of the molecular ion of this compound?

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[A_r values: H = 1.0078 C = 12.0000 O = 15.9949]

- ☒ **A** 92.0261
- ☒ **B** 92.0312
- ☒ **C** 93.0339
- ☒ **D** 93.0390

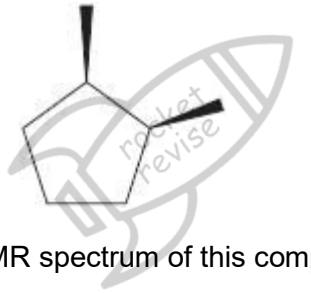
(Total for question = 1 mark)

Q27.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The structure of a hydrocarbon is shown.



How many peaks will there be in the ^{13}C NMR spectrum of this compound?

- ☒ **A** four
- ☒ **B** five
- ☒ **C** six
- ☒ **D** seven

(Total for question = 1 mark)

Q28.

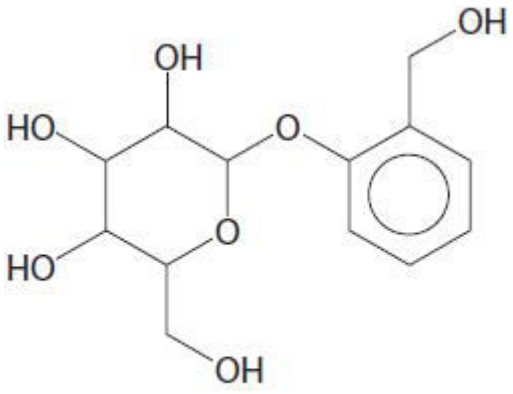
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Chemicals from Plants

Plants are a rich source of useful chemicals, although their applications have often pre-dated the identification of the active compound. One of the best known examples of this is the use of willow bark extracts to reduce pain and fevers, a practice that is at least two thousand years old. The active compound in willow bark is salicin.

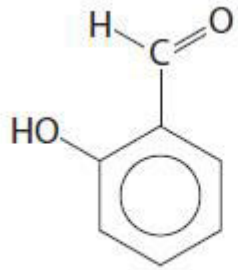
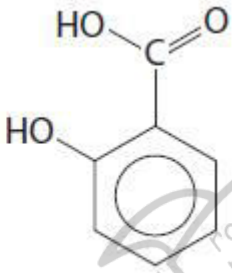
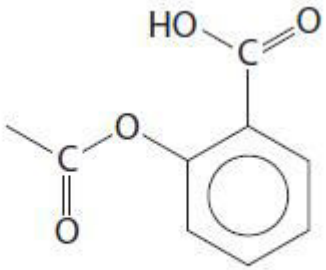
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salicin

In the body, salicin is changed into salicylaldehyde and then salicylic acid. Salicylic acid may in turn be converted into 2-acetoxybenzenecarboxylic acid, a compound which is better known as aspirin, one of the most widely used medications in the world.

		
salicylaldehyde	salicylic acid	2-acetoxybenzenecarboxylic acid

(a) Calculate the percentage composition by mass of the elements in salicin.

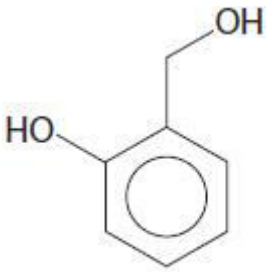
(4)

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(b) The first stage in the breakdown of salicin results in the formation of salicyl alcohol.

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salicyl alcohol

Salicyl alcohol is readily oxidised in the laboratory to form salicylic acid.

(i) State the reagents and conditions needed for this oxidation.

(2)

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(ii) The boiling temperature of salicylaldehyde is 197 °C.
Suggest why this makes it very difficult to obtain salicylaldehyde by oxidising salicyl alcohol.

(2)

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(c) Aromatic aldehydes such as salicylaldehyde may be prepared in the laboratory by electrophilic substitution.

For example, benzaldehyde may be obtained by reacting benzene with a mixture of carbon monoxide and hydrogen chloride in the presence of aluminium chloride.
The mixture of carbon monoxide and hydrogen chloride reacts like methanoyl chloride.

(i) Write an equation for the formation of the electrophile from methanoyl chloride. Use displayed formulae.

(1)

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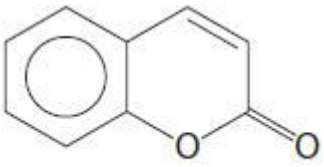
(ii) Draw the mechanism of the formation of benzaldehyde from benzene using the electrophile from (c)(i).

(3)

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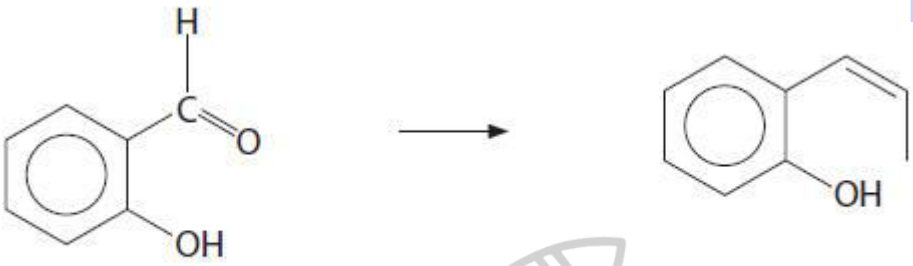
(d) Salicylaldehyde may be used in the synthesis of coumarin, a compound which occurs in many plants. Coumarin is in turn used to prepare warfarin, a compound prescribed to reduce blood clotting.



coumarin

One suggested synthesis of coumarin from salicylaldehyde involves the formation of an intermediate compound, **F**.

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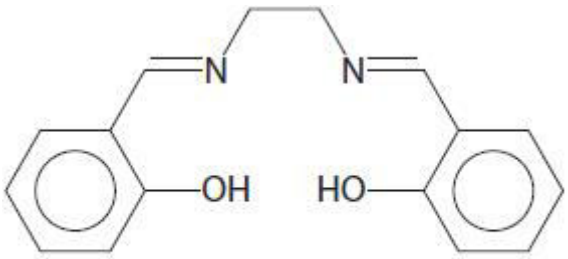
salicylaldehyde compound **F**

Devise a synthesis of **F** using salicylaldehyde and bromoethane as the **only** organic starting materials. Include any other reagents and intermediate compounds, and give essential reaction conditions.

(4)

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(e) Salicylaldehyde combines with 1,2-diaminoethane in a condensation reaction to form salen ligand.



salen ligand

Salen ligand reacts with many metal ions to form very stable complexes which are useful catalysts.

(i) Draw a diagram of the complex that **one** salen ligand forms with a Ni^{2+} ion, showing the type of

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bonding involved.

(2)

(ii) Explain why the salen ligand complex of the Ni^{2+} ion is much more stable than the aqua complex of the same ion.

(2)

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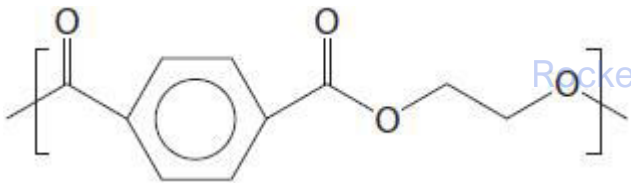
(Total for question = 20 marks)

Q29.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about polymers.

(a) A repeat unit of the polymer PET has the structure shown.



What is the percentage by mass of carbon in the repeat unit?

(1)

- ☐ **A** 57.1 %
- ☐ **B** 62.5 %
- ☐ **C** 65.2 %
- ☐ **D** 66.7 %

(b) One of the monomers used to make PET is benzene-1,4-dicarboxylic acid.

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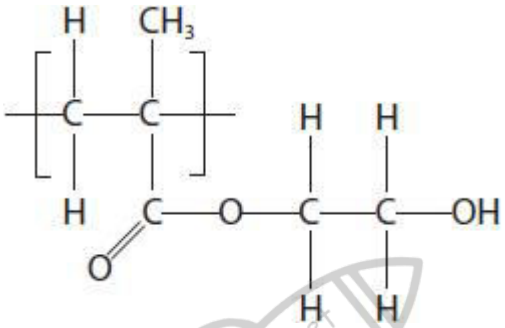


How many peaks are there in the 13C NMR spectrum of benzene-1,4-dicarboxylic acid?

(1)

- ☒ **A** 2
- ☒ **B** 3
- ☒ **C** 4
- ☒ **D** 6

(c) The repeat unit of another polymer has the structure shown.



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This polymer is formed from

(1)

- ☒ **A** a single type of monomer by an addition reaction
- ☒ **B** a single type of monomer by a condensation reaction
- ☒ **C** two different types of monomer by an addition reaction
- ☒ **D** two different types of monomer by a condensation reaction

Use this space for rough working. Anything you write in this space will gain no credit.

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(Total for question = 3 marks)

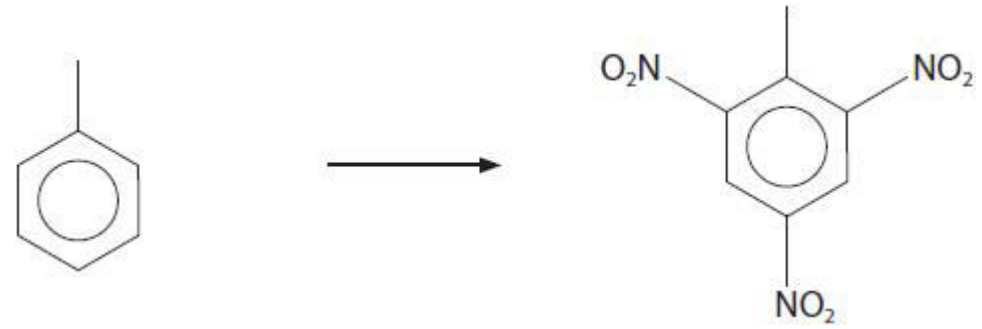
Q30.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Methylbenzene reacts with a mixture of concentrated nitric acid and concentrated sulfuric acid to form 2,4,6-trinitromethylbenzene.



(a) What is the number of peaks in the ^{13}C NMR spectrum of methylbenzene?

(1)

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- ☒ **A** seven
- ☒ **B** six
- ☒ **C** five
- ☒ **D** four

(b) What type of reaction takes place?

(1)

- ☒ **A** nucleophilic addition
- ☒ **B** nucleophilic substitution
- ☒ **C** electrophilic addition
- ☒ **D** electrophilic substitution

(c) Which expression shows the mass in grams of 2,4,6-trinitromethylbenzene formed from 10 g of methylbenzene if the yield of the reaction is 85 %?

[M_r values: methylbenzene = 92 2,4,6-trinitromethylbenzene = 227]

(1)

- ☒ **A** (10 × 85 × 227) ÷ (92 × 100)
- ☒ **B** (10 × 100 × 227) ÷ (92 × 85)
- ☒ **C** (10 × 100 × 227) ÷ (92 × 115)
- ☒ **D** (10 × 115 × 227) ÷ (92 × 100)

(Total for question = 3 marks)

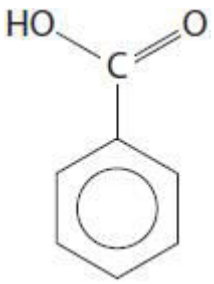
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Q31.

Benzoic acid is a white crystalline solid with the structure shown.

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It is found in many plants as it is an important building block for the biosynthesis of a variety of compounds, such as plant hormones and attractants for pollinators.

The role of benzoic acid in the chemical industry is also widespread and approximately 500 000 tonnes are produced annually. It is used in the synthesis of many compounds, including medicines, dyes and insect repellents.

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Such synthetic dyes are often classified as aryl azo dyes. These dyes have a range of vivid colours and a wide range of uses in many industries, including food and textiles. Their synthesis involves the formation of a diazonium ion. This ion then reacts with a phenol in a coupling reaction, to form the dye. The relative simplicity of the reactions involved and ready availability of starting materials make azo dyes cheap to produce.

Salts of benzoic acid, such as calcium benzoate and sodium benzoate, are used in the food industry as preservatives.

(a) Devise a reaction scheme to produce benzoic acid from benzene, via bromobenzene and then a Grignard reagent.

Include the reagents and essential conditions for each step and give the name or structure of each of the intermediate compounds.
Details of practical procedures and reaction mechanisms are **not** required.

(6)

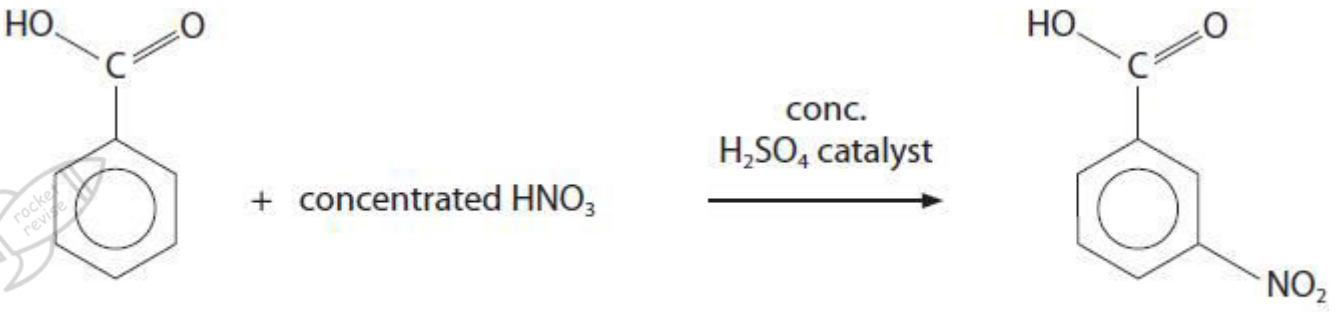
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(b) Benzoic acid can be used in the synthesis of azo dyes.

(i) In Step **1**, benzoic acid reacts with concentrated nitric acid to form 3-nitrobenzoic acid.

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benzoic acid

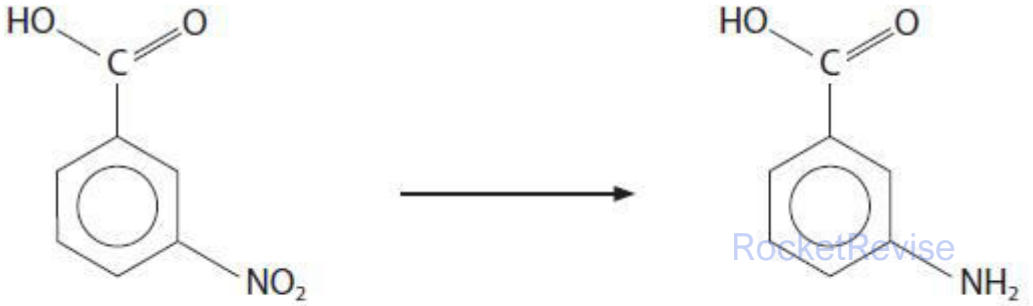
3-nitrobenzoic acid

Draw the mechanism for the reaction, using appropriate curly arrows.
Include equations showing the role of the catalyst and how it is regenerated.

(5)

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(ii) In Step **2**, the 3-nitrobenzoic acid reacts to form 3-aminobenzoic acid.



3-nitrobenzoic acid

3-aminobenzoic acid

State the reagents required for this reaction.

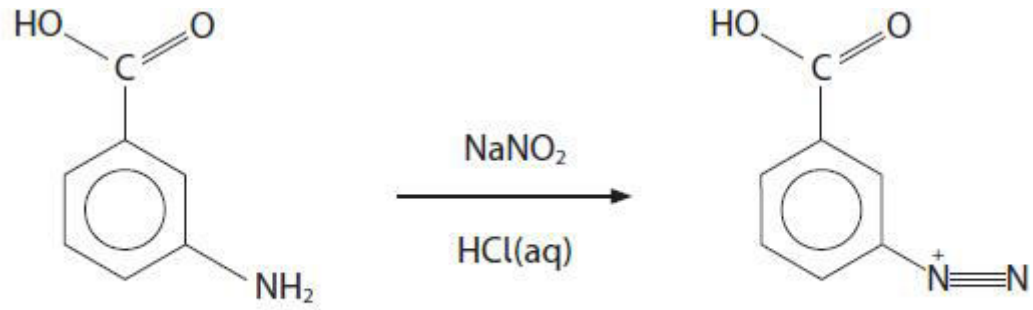
(1)

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(iii) In Step **3**, the 3-aminobenzoic acid reacts with sodium nitrite and dilute hydrochloric acid, forming a diazonium ion.

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3-aminobenzoic acid diazonium ion

State a temperature at which this reaction should take place, giving **one** reason for your answer.

(2)

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(iv) Draw the structure of the azo dye formed when the diazonium ion reacts with phenol.

(1)

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(c) Hydrated calcium benzoate is used as a preservative in soft drinks. It has the formula $Ca(C_6H_5COO)_2 \cdot xH_2O$.

2.60 g of hydrated calcium benzoate was dissolved in deionised water. Excess lead(II) nitrate solution was added, forming a precipitate of lead(II) benzoate, $Pb(C_6H_5COO)_2(s)$. This precipitate was removed and dried. The mass of the dry solid was 3.89 g.

Calculate the molar mass of hydrated calcium benzoate and hence deduce the value of x.

(6)

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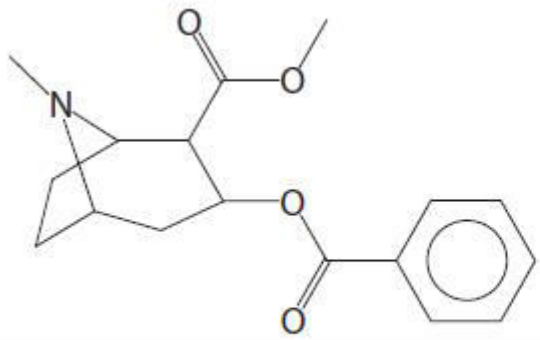
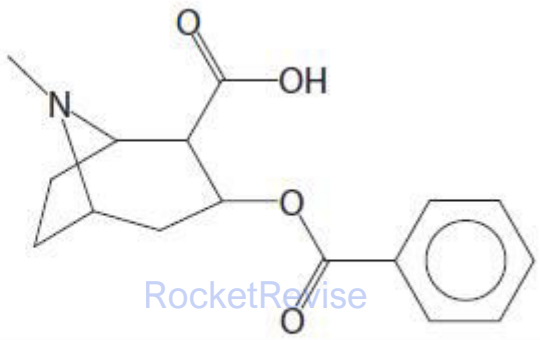
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(Total for question = 21 marks)

Q32.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The structures of cocaine and its metabolite benzoylecgonine are shown.

	
cocaine	benzoylecgonine

How would you expect the solubility of cocaine in water and the pH of its aqueous solution to compare with benzoylecgonine?

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		Solubility in water	pH of aqueous solution
☒	A	cocaine more soluble	cocaine higher pH
☒	B	cocaine more soluble	cocaine lower pH
☒	C	cocaine less soluble	cocaine higher pH
☒	D	cocaine less soluble	cocaine lower pH

(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

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Q33.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The use of recrystallisation to purify a chemical compound depends on how its solubility in the chosen solvent varies with temperature.

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How should the solubility of the chemical compound depend on temperature?

		High temperature	Low temperature
☒	A	soluble	soluble
☒	B	soluble	insoluble
☒	C	insoluble	soluble
☒	D	insoluble	insoluble

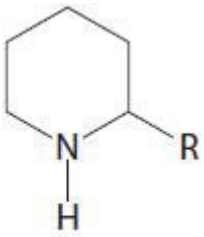
(Total for question = 1 mark)

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Q34.

Coniine is the toxic compound present in poison hemlock.
The structure of coniine is shown, with R representing an alkyl group.



coniine

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(a) A sample of 0.235 g of coniine was vaporised at 185 °C and 105 000 Pa.

The volume of the vapour was 67.1 cm^3.

(i) Show by calculation that the molar mass of coniine is 127g mol^{-1}.

(4)

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(ii) Deduce the molecular formula of the alkyl group R, using the structure of coniine and its molar mass. You **must** show your working.

(2)

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(b) The table summarises the low resolution proton NMR data for the R group in coniine.

Proton environment	Chemical shift / ppm	Peak area
1	0.90	3
2	1.33	2
3	1.37	2

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(i) Explain why only **one** of the two possible structural formulae of R can give these data.

(3)

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(ii) In the high resolution proton NMR data for the R group in coniine, the peak for proton environment 2 is a sextet.

Deduce the splitting patterns for proton environments 1 and 3, using this information and the information in the table.

(3)

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(c) Explain, using a diagram, which type of stereoisomerism is shown by coniine. In your diagram use R to represent the alkyl group.

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(2)

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(Total for question = 14 marks)

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Q35.

This question is about benzene and some of its compounds.

(a) Benzene reacts with bromine in the presence of an iron(III) bromide catalyst to form bromobenzene.



(i) Draw the mechanism for this reaction, including the formation of the electrophile.

(4)

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* (ii) Compare and contrast the reaction of benzene with bromine with the reactions of cyclohexene with bromine and of phenol with bromine.
Include reasons for any differences.
Detailed mechanisms for these reactions are **not** required.

(6)

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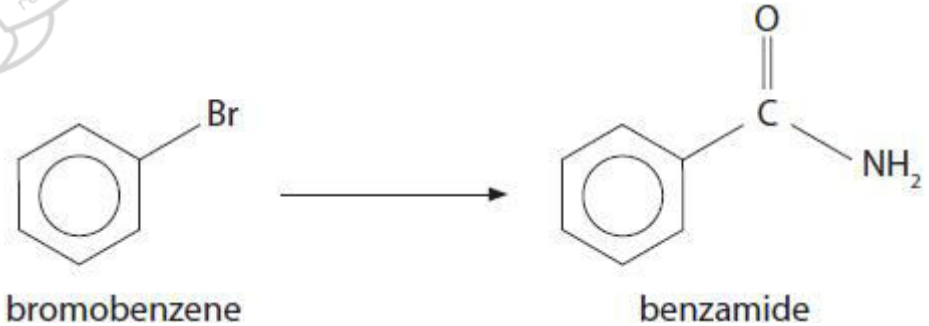
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(b) Devise a synthesis to convert bromobenzene into benzamide.

Include reagents, conditions and the structures of the intermediate compounds.



bromobenzene benzamide

(7)

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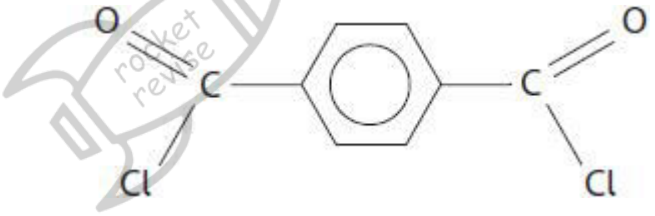
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(c) Semi-aromatic polyamides have high melting temperatures, high strength and rigidity and are resistant to chemical attack so they have many applications in engineering.

(i) Draw the repeat unit of the polyamide formed from the monomers shown.

$H_2N(CH_2)_6NH_2$ and



(1)

(ii) Explain why polyamides have higher melting temperatures than polyalkenes.

(2)

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(Total for question = 20 marks)

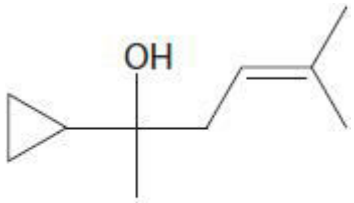
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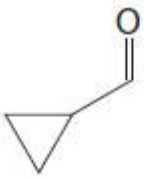
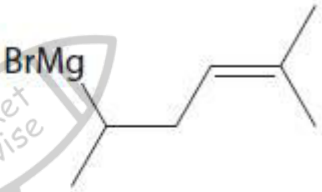
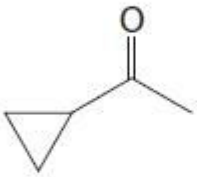
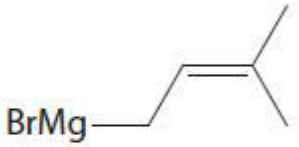
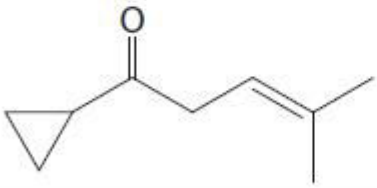
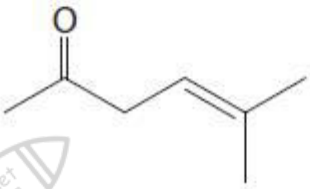
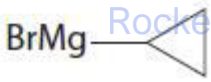
Q36.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which carbonyl compound and Grignard reagent would **not** react to form compound **Y**?



compound **Y**

	Carbonyl compound	Grignard reagent
☐ **A**		
☐ **B**		
☐ **C**		$BrMgCH_3$
☐ **D**		

(Total for question = 1 mark)

Q37.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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When 10 cm^3 of but-1-ene, C_4H_8, is ignited with excess oxygen, there is an overall decrease in volume of x cm^3.

All gas volumes are measured at room temperature and pressure.

$$C_4H_8(g) + 6O_2(g) \rightarrow 4CO_2(g) + 4H_2O(l)$$

A further decrease of y cm^3 takes place on the addition of aqueous sodium hydroxide as carbon dioxide is removed.

What are the values of x and y?

		Value of x	Value of y
☐	**A**	30	40
☐	**B**	30	80
☐	**C**	40	40
☐	**D**	40	80

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(Total for question = 1 mark)

Q38.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Grignard reagents are used to increase the length of the carbon chain in organic molecules.

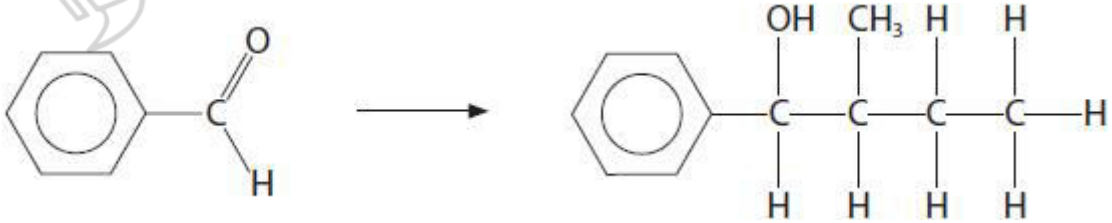
(a) What type of reaction occurs with a Grignard reagent?

(1)

☐ **A** electrophilic addition
☐ **B** free radical substitution
☐ **C** oxidation
☐ **D** nucleophilic addition

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(b) Benzaldehyde reacts with a Grignard reagent, followed by hydrolysis, to form 2-methyl-1-phenylbutan-1-ol.



What is the formula of this Grignard reagent?

(1)

☐ **A** $CH_3CH_2CH_2CH_2MgBr$

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☒ **B** $CH_3CH_2CH(CH_3)MgBr$
☒ **C** $(CH_3)_2CHCH_2MgBr$
☒ **D** $(CH_3)_3CMgBr$

(Total for question = 2 marks)

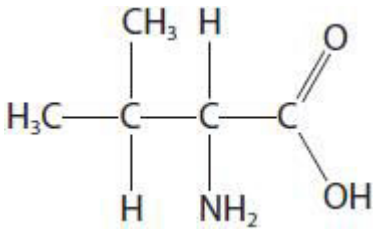
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Q39.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about the infrared and proton NMR spectra of the amino acid valine.



(a) Which of these wavenumber ranges would you **not** expect to have an absorption in the infrared spectrum of valine?

(1)

☒ **A** 3500 – 3300 cm^{-1}
☒ **B** 3095 – 3010 cm^{-1}
☒ **C** 2962 – 2853 cm^{-1}
☒ **D** 1725 – 1700 cm^{-1}

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(b) What are the expected splitting patterns in the high resolution proton NMR spectrum of valine for the protons shown in bold?

(1)

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	Splitting pattern for CH_3 protons	Splitting pattern for $(CH_3)_2CH$ proton
☐ A	doublet	octet
☐ B	doublet	septet
☐ C	singlet	octet
☐ D	singlet	septet

(Total for question = 2 marks)

RocketRevise

Q40.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An organic compound, **J**, reacts with

- sodium hydroxide to form an ionic compound
- hydrogen in the presence of a nickel catalyst
- ethanol in the presence of sulfuric acid.

Which of these could be the structure of **J**?

☐ **A** $CH_2=CHCOOH$

☐ **B** C_6H_5OH

☐ **C** $CH_2=CHCH_2OH$

☐ **D** CH_3COOH

Use this space for any rough working. Anything you write in this space will gain no credit.

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(Total for question = 1 mark)

Q41.

Organic compounds can be analysed using a combination of qualitative analysis, quantitative analysis

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and spectroscopic techniques.

(a) An organic compound **A** contains carbon, hydrogen and oxygen only and has a molar mass of 136 g mol^{-1}.

Complete combustion of 5.26 g of **A** in dry oxygen produces 15.3 g of carbon dioxide and 4.18 g of water.
Show that the formula of **A** is $C_9H_{12}O$.
You **must** show all your working.

(4)

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(b) Compound **A**, $C_9H_{12}O$, contains a benzene ring and **one** side-chain.

A is oxidised when heated with acidified potassium dichromate(VI) to produce a compound **B** which reacts with iodine and aqueous sodium hydroxide to give a pale yellow solid, **C**.
When **A** reacts with concentrated phosphoric(V) acid, it forms a mixture of two **structural isomers**, **D** and **E**, both of which decolourise bromine water.
D exists as geometric isomers but **E** does not.
Draw the structures of compounds **A** to **E**.

(5)

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Structure
A
B
C
D
E

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(c) Compound **F** is a structural isomer of **A** which also has a benzene ring and one side-chain.

F is **not** oxidised when heated with acidified potassium dichromate(VI). The ^{13}C NMR spectrum of **F** has six peaks.

Draw the structure of **F**. Justify your answer by labelling the six different carbon environments.

(2)

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(d) Compound **G** is a structural isomer of **A** which also has a benzene ring and one side-chain.

The mass spectrum of **G** has a peak at $m / z = 107$.
Deduce the structure of **G**. Justify your answer.

(2)

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.....

.....

.....

(Total for question = 13 marks)

Q42.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The organic compound 4-aminophenol can behave as an acid and a base.



What is formed when 4-aminophenol reacts with hydrochloric acid?

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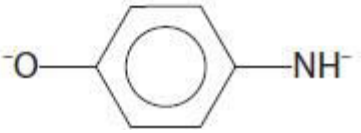
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☒ **A** 

☒ **B** 

☒ **C** 

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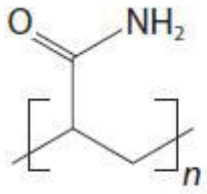
☒ **D** 

(Total for question = 1 mark)

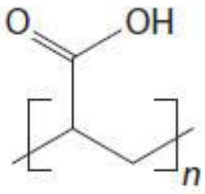
Q43.

This question is about polymers.

(a) Polyacrylamide (PAM) and polyacrylic acid (PAA) are water-absorbent addition polymers.



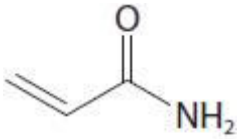
PAM



PAA

RocketRevise

(i) PAM is made from the acrylamide monomer.



acrylamide

Give the IUPAC name for acrylamide.

(1)

.....

(ii) Explain why PAM is able to absorb **large** amounts of water.
Include a diagram in your answer.

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(3)

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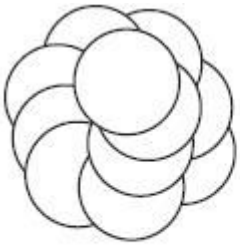
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(iii) The structure of PAA in aqueous solution is pH-dependent.
Below pH 4, the structure of PAA is compact due to the formation of intramolecular hydrogen bonds.
Above pH 8, PAA has an open coil structure.



compact structure
below pH 4



open coil structure
above pH 8

Suggest why PAA exists as an open coil structure above pH 8.

(2)

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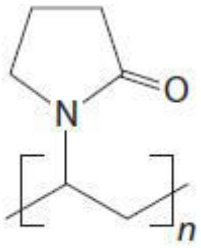
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(b) Polyvinylpyrrolidone (PVP) is an addition polymer used in the pharmaceutical industry. The average molar mass of PVP is 90 000 g mol^{-1}.

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PVP

(i) Draw the structure of the vinylpyrrolidone monomer used to make PVP.

(1)

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(ii) Calculate the number of monomers needed to make one molecule of PVP polymer with a molar mass of 90 000 g mol^{-1}.
Give your answer to the nearest whole number.

(2)

(iii) A 740 mg tablet of a painkiller contains 4.0 % PVP by mass.
Calculate the number of molecules of PVP polymer in the tablet.

(3)

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(c) All types of nylon are condensation polymers.

(i) State what is meant by the term **condensation polymer**.

(2)

.....

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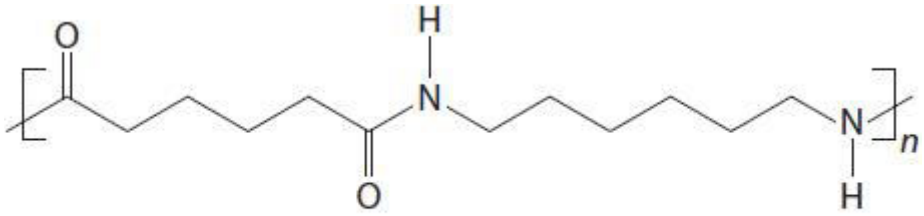
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(ii) The structure of nylon 6,6 is shown.



nylon 6,6

Nylon 6,6 is made from two monomers.
Deduce the **structural** formulae of these two monomers.

(2)

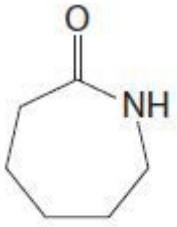
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structural formula of monomer 1

structural formula of monomer 2

(iii) Caprolactam can be directly converted to give a different polymer, nylon 6, in a ring-opening polymerisation reaction.

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caprolactam

Draw **two** repeat units of nylon 6.

(2)

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(Total for question = 18 marks)

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Q44.

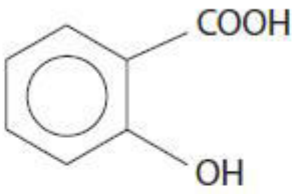
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Salicylic acid

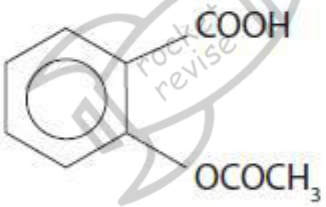
Salicin was extracted from the bark of willow trees in the first half of the 19th century. About 30 years later, scientists showed that salicin was converted into salicylic acid in the body.



salicylic acid

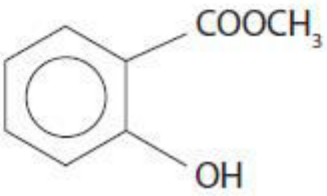
Salicylic acid was given to patients with fevers and their symptoms were reduced. However, it caused severe irritation to the lining of the mouth, oesophagus and stomach.

In the 1890s Hoffmann converted salicylic acid into acetylsalicylic acid, which is commonly known as aspirin and is still used to treat fevers and as a painkiller.



acetylsalicylic acid

Salicylic acid may also be converted into methyl salicylate. This is the active ingredient in oil of wintergreen, which is applied externally to treat joint and muscle pain.



methyl salicylate

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(a) Give the IUPAC name for salicylic acid.

(1)

.....

(b) Acetylsalicylic acid is formed when salicylic acid reacts with ethanoic anhydride, $(CH_3CO)_2O$.

(i) Name the three functional groups in acetylsalicylic acid.

(1)

.....

(ii) Write the equation for the reaction of salicylic acid and ethanoic anhydride.

(1)

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(iii) 2.00 g of salicylic acid produced a 74.8% yield of acetylsalicylic acid.
Calculate the mass of acetylsalicylic acid formed.
[M_r values: salicylic acid = 138 acetylsalicylic acid = 180]

(3)

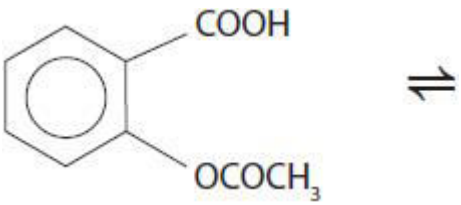
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(c) Acetylsalicylic acid is only slightly soluble in water.

It has a K_a value of 2.8×10^{-4} mol dm^{-3}.

(i) Complete the equation to show the dissociation of acetylsalicylic acid in aqueous solution.

(1)



(ii) The dissociation of acetylsalicylic acid is different in the acidic conditions found in the stomach compared with the slightly alkaline conditions in the small intestine.
Explain this difference in dissociation.

(3)

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(d) Identify, by name or formula, the reagent needed to convert salicylic acid into methyl salicylate.

(1)

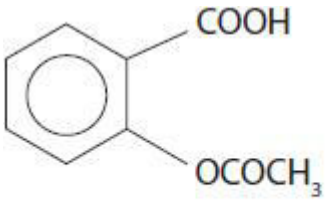
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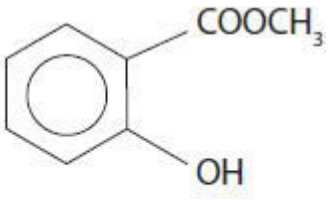
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(e) The high resolution proton (^1H) NMR spectra of acetylsalicylic acid and methyl salicylate both contain four peaks from the protons on the benzene ring and two other peaks from the OH and CH_3 protons in the side chains.

Complete the table to give the chemical shift ranges that you would expect for each type of proton in the side chains of the molecules.



acetylsalicylic acid



methyl salicylate

(2)

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	Acetylsalicylic acid		Methyl salicylate	
Type of proton	OH	CH_3	OH	CH_3
Chemical shift / ppm				

(f) Acetylsalicylic acid in aspirin tablets is hydrolysed by excess sodium hydroxide.

$$CH_3COOC_6H_4COOH + 2NaOH \rightarrow CH_3COONa + HOC_6H_4COONa + H_2O$$

The unreacted sodium hydroxide may be titrated with a standard solution of hydrochloric acid to determine the percentage of acetylsalicylic acid in the tablets.
Aspirin tablets can be analysed using the outline procedure:

- add 25.0 cm^3 of 1.00 mol dm^{-3} sodium hydroxide to a known mass of powdered aspirin tablets
- heat the mixture for 10 minutes to hydrolyse the aspirin
- cool the mixture, make the solution up to 250.0 cm^3 in a volumetric flask and mix it thoroughly
- pipette 25.0 cm^3 of the hydrolysed solution into a conical flask and titrate it against 0.100 mol dm^{-3} hydrochloric acid using phenolphthalein indicator
- repeat the titration until concordant values are obtained.

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Results
Mass of aspirin tablets used = 0.760 g
Mean titre = 16.95 cm^3
The table shows the percentage by mass of acetylsalicylic acid in four brands of aspirin tablets.

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Brand of aspirin tablets	Percentage of acetylsalicylic acid / %
A	92.2
B	95.4
C	97.5
D	99.6

Determine which brand of aspirin tablets was analysed.
You **must** show your working.

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(6)

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(Total for question = 19 marks)

Q45.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A 10 cm^3 mixture of methane and argon is ignited.
The methane requires 4 cm^3 of oxygen for complete combustion.

What is the volume of **argon**, in cm^3, in the mixture?

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All volumes are measured at the same temperature and pressure.

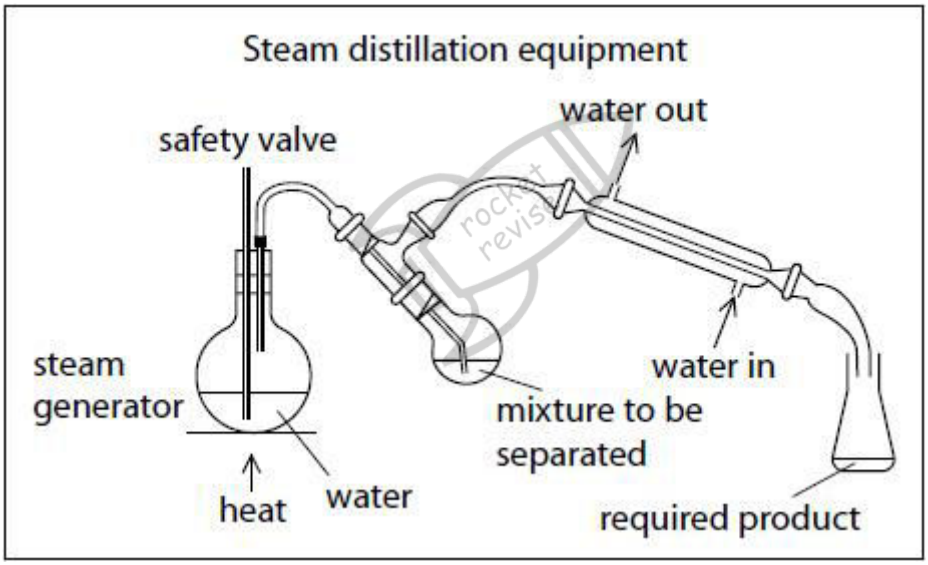
- **A** 2
- **B** 4
- **C** 6
- **D** 8

(Total for question = 1 mark)

Q46.

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Steam distillation is a technique used during some organic preparations to separate the product from the reaction mixture.



What benefit is gained from the use of steam distillation compared to other methods of distillation?

(1)

- **A** a pure distillate is produced
- **B** high distillation temperatures are required
- **C** it works well for molecules miscible with water
- **D** it avoids the decomposition of the organic molecule when it distils

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(Total for question = 1 mark)

Q47.

Butan-2-ol is a secondary alcohol with four carbon atoms.

(a) Devise a reaction scheme to form butan-2-ol from iodoethane, C_2H_5I, as the only organic compound. Give reagents, conditions and equations for each of the steps.

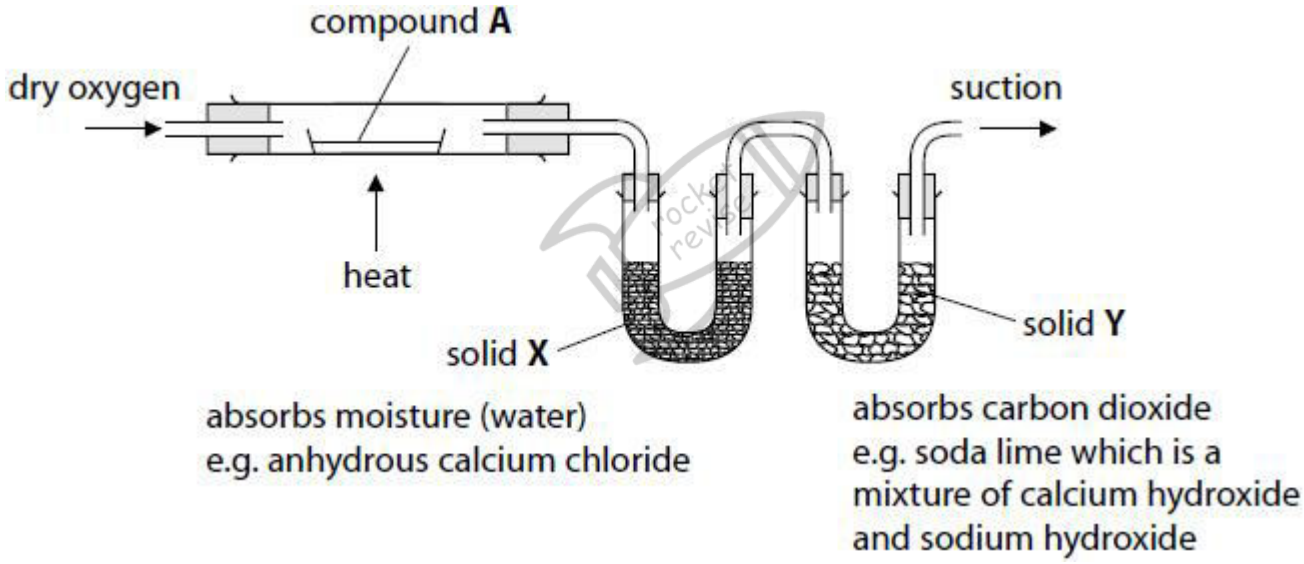
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(8)

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(b) A 1.850 g sample of an organic substance, compound A, that is thought to be butan-2-ol is tested by combustion analysis using the apparatus shown.



(i) Calculate the mass increase of solid X and solid Y that would result if compound A is butan-2-ol.

(4)

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(ii) Predict a substance which would give the same mass increase in solids X and Y from combustion analysis as butan-2-ol.

Give a reason for your prediction.

(2)

.....

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(Total for question = 14 marks)

Q48.

What is the number of peaks in a ^{13}C NMR spectrum of 1,4-dimethylbenzene?



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(1)

- ☒ **A** 3
- ☒ **B** 4
- ☒ **C** 7
- ☒ **D** 8

(Total for question = 1 mark)

Q49.

This question is about the investigation of an organic compound **X**.
X is a liquid at room temperature and pressure, which turns damp red litmus paper blue.

(a) (i) Name the functional group present in **X**.

(1)

.....

(ii) When 0.493 g of **X** was vaporised, 157 cm^3 of dry air was displaced, measured at 15°C and 103 000 Pa.
Calculate the molar mass of **X**, using the ideal gas equation.
You **must** show your working.

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(4)

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(b) **X** reacted vigorously with ethanoyl chloride forming steamy fumes and a white solid **Y**.

(i) Identify the steamy fumes, by name or formula.

(1)

.....

.....

(ii) Suggest the functional group present in **Y**.

(1)

.....

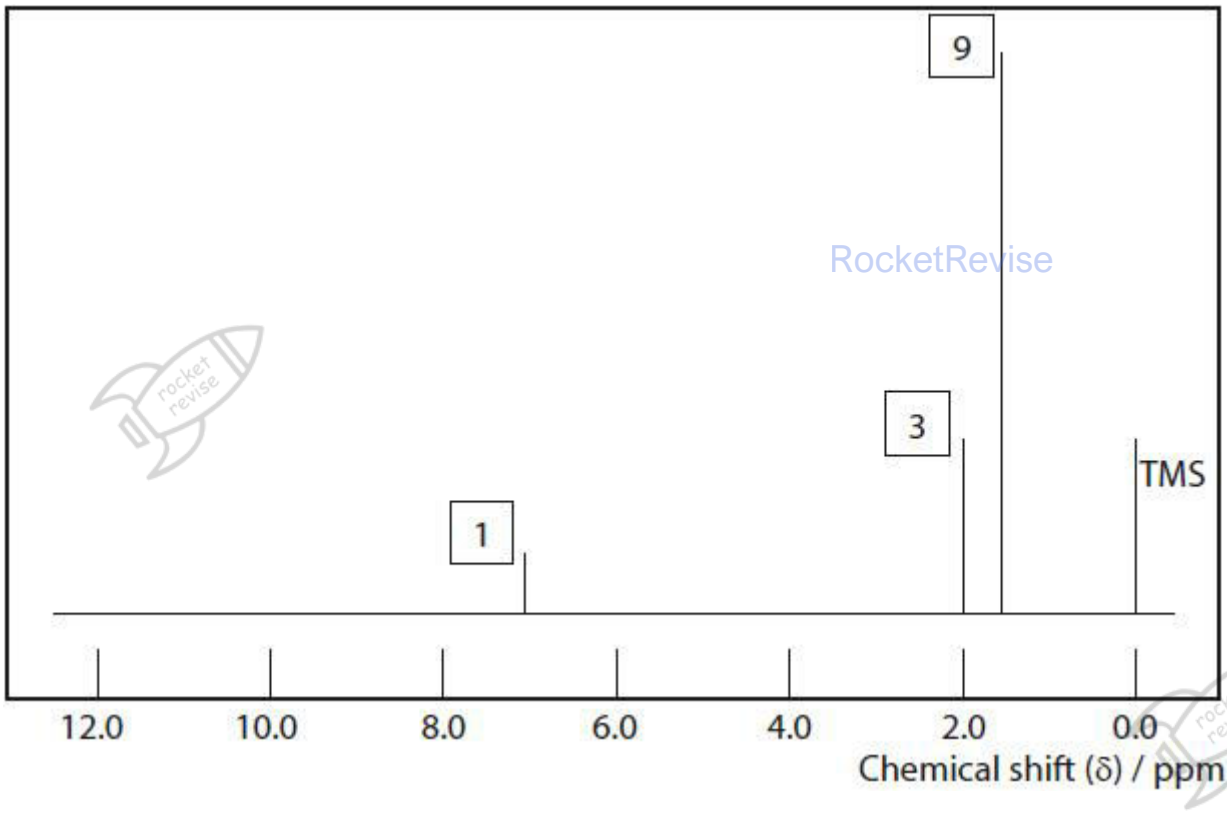
(iii) Analysis of **Y** showed that its composition by mass was 62.6 % carbon; 11.3 % hydrogen; 12.2 % nitrogen; 13.9 % oxygen.

Determine the empirical formula of **Y**. You **must** show your working.

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(3)

*(c) A simplified **high** resolution proton NMR spectrum of **Y** is shown.
The relative peak areas are given near each peak.



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Deduce the structure of **Y**, using the NMR spectrum and the other information in the question.

(6)

(d) Draw the structure of compound **X**.

(1)

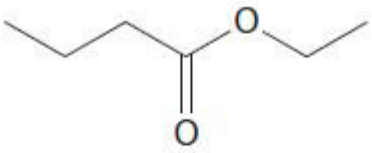
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(Total for question = 17 marks)

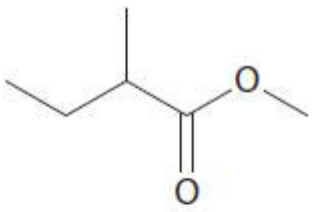
Q50.

Answer the question with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

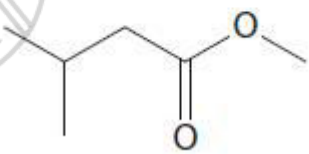
The aroma of strawberries is due to a number of volatile compounds, including the four isomeric esters shown.



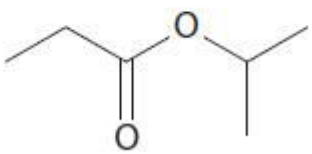
ester **P**



ester **Q**



ester **R**



ester **S**

(a) Which of the esters have five peaks in their ^{13}C NMR spectrum?

(1)

- ☒ **A** **P** only
- ☒ **B** **Q** and **R** only
- ☒ **C** **R** and **S** only

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☒ **D** **Q**, **R** and **S** only

(b) Which of the esters will **not** have a doublet in its high resolution proton NMR spectrum?

(1)

☒ **A** **P**
☒ **B** **Q**
☒ **C** **R**
☒ **D** **S**

(c) Which of the esters could rotate the plane of plane–polarised monochromatic light?

(1)

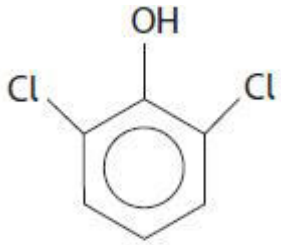
☒ **A** **Q** only
☒ **B** **Q** and **R**
☒ **C** **Q**, **R** and **S**
☒ **D** **R** and **S**

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(Total for question = 3 marks)

Q51.

2,6-dichlorophenol is used for communication between ticks, small parasites that can infect animals including humans.



2,6-dichlorophenol

The compound can be synthesised in the laboratory from phenol.

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4â€‘phenolsulfonic acid is formed in Reaction **A**. This is then chlorinated.

Hydrolysis of intermediate compound **B** removes the sulfonic acid group.

The reaction scheme is shown.

phenol $\xrightarrow[\text{A}]{\text{Reaction}}$ 4-phenolsulfonic acid $C_6H_6O_4S$ $\xrightarrow{Cl_2}$ Intermediate **B** $\xrightarrow{\text{Hydrolysis}}$ 2,6-dichlorophenol

(a) (i) The sulfonation of phenol is similar to the sulfonation of benzene.

Suggest the reagent(s) required for Reaction **A**.

(1)

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(ii) Deduce the structure of intermediate compound **B**.

(1)

(iii) Explain why the reaction of chlorine with phenol occurs under milder conditions than the reaction of chlorine with benzene.

(3)

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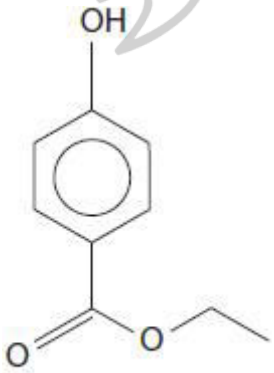
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(b) An alternative synthesis of 2,6-dichlorophenol starts with compound **C**.



compound **C**

Compound C is chlorinated and hydrolysed and then the CO_2 group is removed (decarboxylation) to form 2,6-dichlorophenol.
Other than phenol, name the functional group present in compound **C**.

(1)

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(c) Identify **three** factors that organic chemists would take into account when considering alternative methods for an organic synthesis.

(3)

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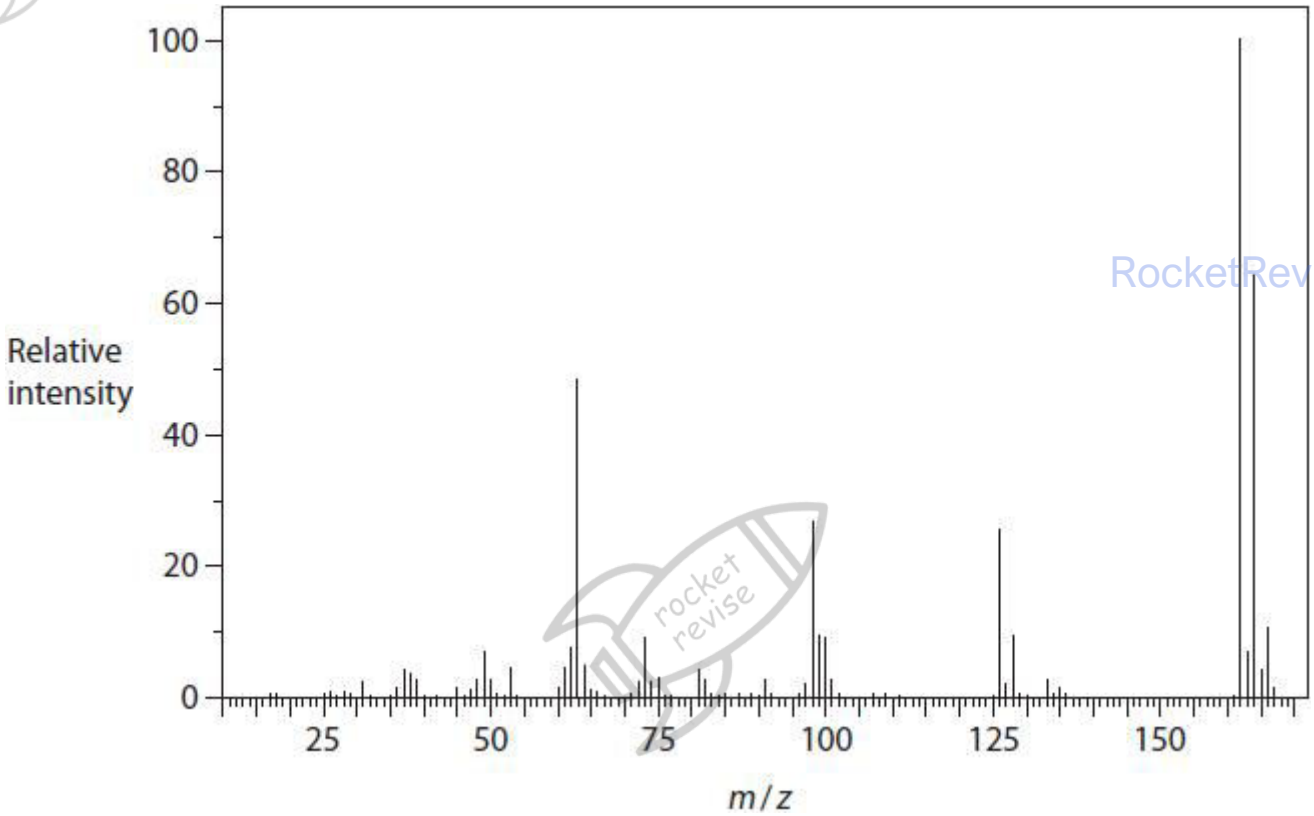
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(d) The mass spectrum of 2,6-dichlorophenol is shown.



(i) Explain the relative intensities of the peaks at *m/z* values of 162, 164 and 166.

(2)

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(ii) Suggest why there is a corresponding set of peaks at *m/z* values of 163, 165 and 167.

(1)

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(Total for question = 12 marks)

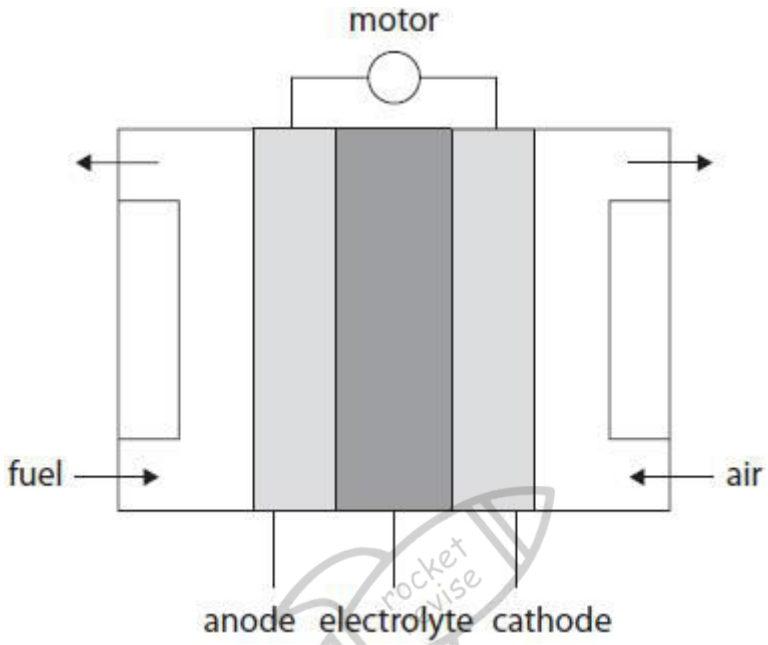
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Q52.

*Fuel cells are an alternative to internal combustion engines and batteries for powering vehicles.

A schematic diagram of a fuel cell is shown.



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Explain how a fuel cell works, discussing their advantages and disadvantages.

Use the acidic hydrogen-oxygen fuel cell as an example.

Include, in your answer, half-equations for the electrode reactions and an example of another fuel suitable for use in fuel cells.

$E^{\ominus}$ values and $E^{\ominus}_{cell}$ values are not required.

(6)

(Total for question = 6 marks)

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Q53.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Grignard reagents are very useful for increasing the length of the carbon chain of organic compounds.

(a) Which metal is involved in the formation of a Grignard reagent?

(1)

- ☐ **A** chromium
- ☐ **B** copper
- ☐ **C** magnesium

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☒ **D** manganese

(b) What best describes the role of a Grignard reagent in increasing the length of a carbon chain?

(1)

☒ **A** reducing agent
☒ **B** oxidising agent
☒ **C** nucleophile
☒ **D** electrophile

(Total for question = 2 marks)

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Q54.

In recent years, there have been health and safety concerns over "liquitabs". These are capsules which contain concentrated liquid detergents.

The soluble packaging breaks down upon contact with water during a laundry cycle to release the detergent.

Some liquitabs are small and colourful, which makes them attractive to young children who may be tempted to put them in their mouth or play with them. When moist, liquitabs can burst in a child's hand, potentially irritating the eyes and skin. They can also dissolve quickly and may burst in a child's mouth in a very short period of time. This allows the contents to escape quickly and if swallowed can cause severe breathing difficulties.

(a) Identify the hazard symbol(s) that should be displayed on a box of liquitabs, using ticks (✓).

(1)

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(b) The capsule containing the detergent is often made from poly(ethenol) and its derivatives.

(i) Draw the structure of poly(ethenol), showing three repeat units.

(2)

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(ii) Use your structure from (b)(i) to calculate the percentage of hydroxyl groups by mass in the polymer.

(2)

(iii) Explain why poly(ethenol) is soluble in water using your answers from (b)(i) and (ii).

(2)

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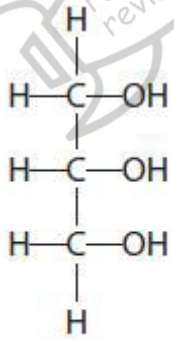
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(c) In liquitab detergents, glycerol is added to the poly(ethenol).



glycerol

(i) Give the IUPAC name for this molecule.

(1)

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(ii) Suggest **one** way that the properties of poly(ethenol) will be improved by the use of glycerol, apart from any effect on solubility.

(1)

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(iii) Glycerol is added to the polymer at 0.110%, by volume.
Calculate the number of moles of glycerol added to 1 tonne of poly(ethenol).

Data: 1 tonne = 1000 kg

density of glycerol: 1.26 g cm^{-3}

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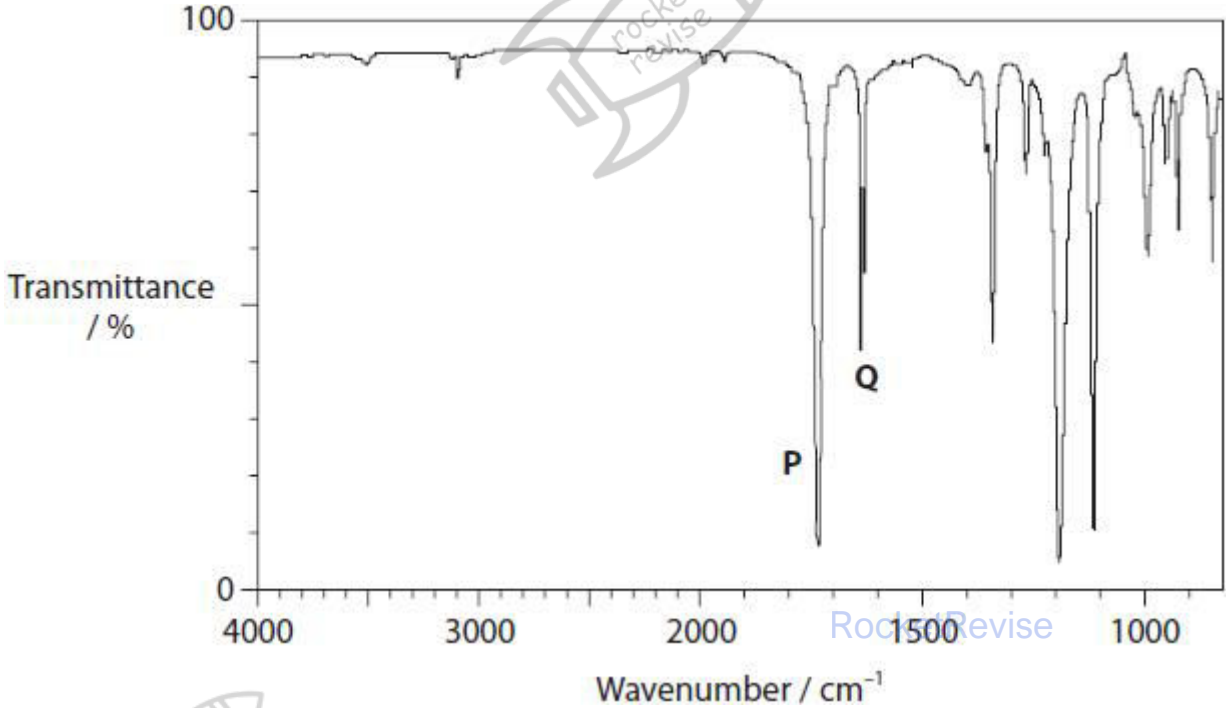
density of poly(ethenol): 1.19 g cm^{-3}

(4)

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(d) Poly(ethenol) is made from a polymer intermediate, rather than by polymerisation, as ethenol is unstable.

The polymer intermediate is prepared from a monomer, **X**.
When 1 mol of **X** is burned in excess oxygen, 95.6 dm^3 of carbon dioxide (at r.t.p.) and 54.1 g of water are formed.
The infrared spectrum for the monomer is shown.



Use the information to deduce the displayed structure of the monomer of the polymer intermediate.
You must show your working and quote the relevant wavenumber ranges from your Data Booklet.

(5)

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(e) Some hospitals use poly(ethenol) laundry bags instead of reusable, washable polyester or nylon bags.

State **two** advantages of using soluble poly(ethenol) laundry bags.

(2)

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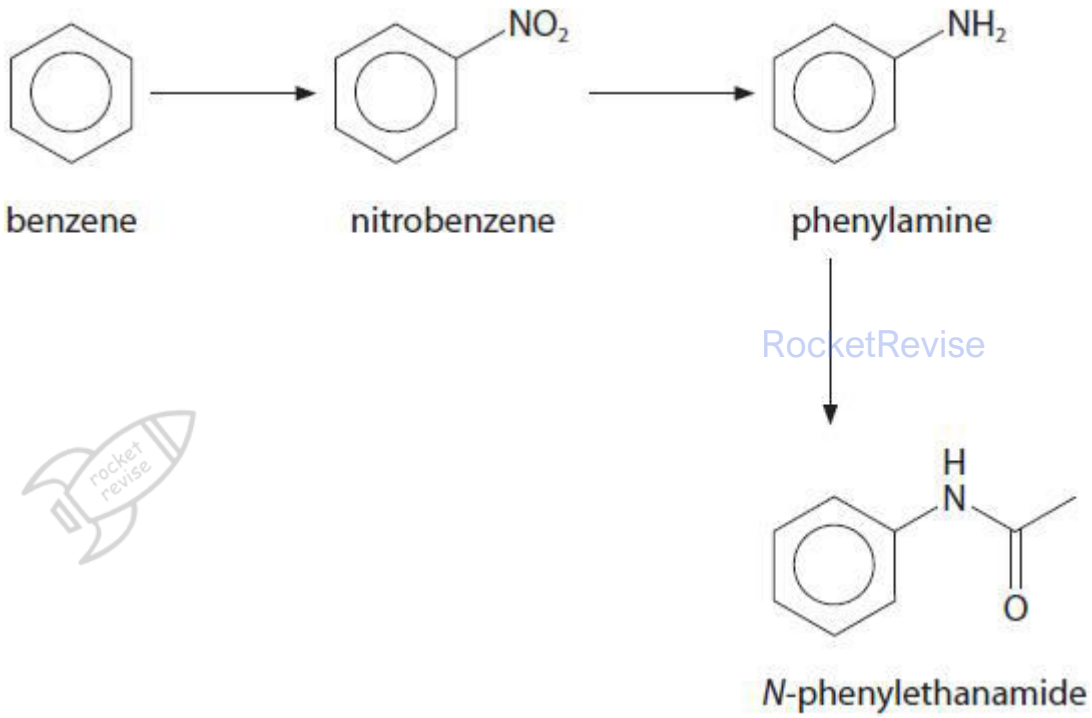
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(Total for question = 20 marks)

Q55.

N-phenylethanamide, historically used as a painkiller, can be synthesised from benzene as shown.



(a) Concentrated nitric acid reacts with a second reagent to produce an electrophile.

This electrophile reacts with benzene to form nitrobenzene.

(i) Identify, by name or formula, the second reagent and the electrophile.

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(2)

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(ii) Draw the mechanism for the reaction between the electrophile and benzene to form nitrobenzene.

(3)

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(b) (i) Identify, by name or formula, the reagent(s) required to convert nitrobenzene into phenylamine.

(1)

.....

(ii) State the type of reaction occurring during this step.

(1)

.....

*(c) In the final step of the synthesis, phenylamine reacts with either ethanoyl chloride or ethanoic anhydride.

phenylamine + ethanoyl chloride → *N*-phenylethanamide + hydrogen chloride
phenylamine + ethanoic anhydride → *N*-phenylethanamide + ethanoic acid

Ethanoyl chloride is considerably more reactive than ethanoic anhydride.

Hazard symbols for reactants and products are shown.

Compound	Hazards
phenylamine	
ethanoic anhydride	
ethanoyl chloride	

Compound	Hazards
N-phenylethanamide	
ethanoic acid	
hydrogen chloride	

Assess the advantages and disadvantages of the use of ethanoic anhydride rather than ethanoyl chloride for this reaction.

Consider the hazards associated with the reactants and products, and the atom economy of each reaction.

(6)

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(d) The overall yield for the synthesis of *N*-phenylethanamide from benzene was found to be 35.2%.

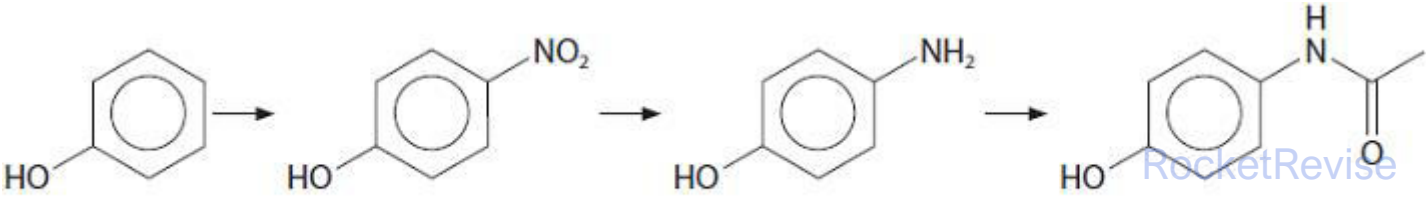
Calculate the minimum volume of benzene, in cm^3, required to make 10.0 g of *N*-phenylethanamide.
[Density of benzene = 0.879 g cm^{-3}]

(4)

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(e) Another painkiller, paracetamol, can be synthesised from phenol in a similar sequence. Phenol is nitrated by dilute nitric acid.



phenol 4-nitrophenol 4-aminophenol paracetamol

Explain why the nitration of phenol requires much milder conditions than the formation of nitrobenzene.

(2)

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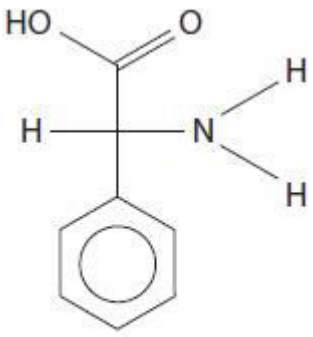
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(Total for question = 19 marks)

Q56.

Phenylglycine is an amino acid which can be synthesised from benzene.



phenylglycine

(a) The first step of the synthesis is the formation of methylbenzene, from benzene and chloromethane.

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Draw the mechanism of this reaction.
Equations for the formation of the electrophile and regeneration of the catalyst are shown.

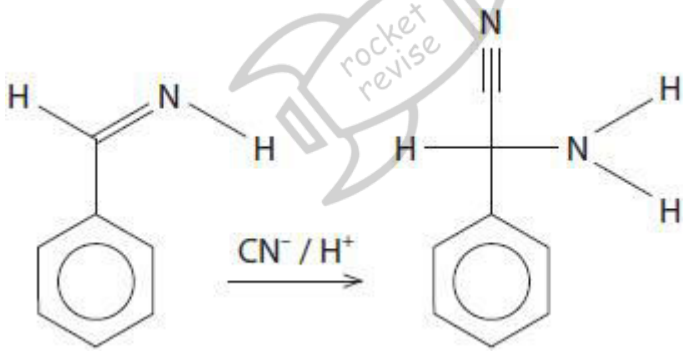
$$CH_3Cl + AlCl_3 \rightarrow CH_3^+ + AlCl_4^-$$

$$AlCl_4^- + H^+ \rightarrow AlCl_3 + HCl$$

(3)

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(b) One of the steps in the synthesis is shown.



Explain, with reference to the mechanism, why this step in the reaction sequence will not produce a single isomer.

(3)

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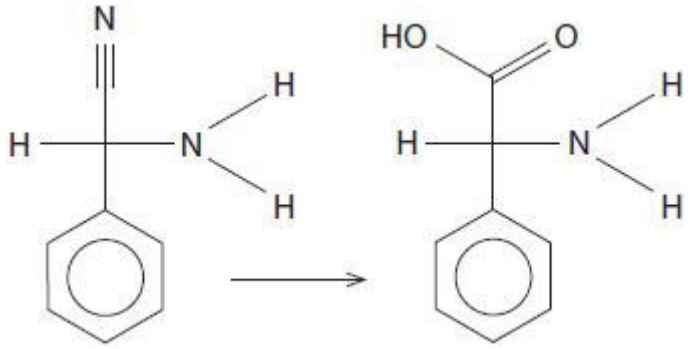
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(c) The final step in the synthesis is shown.

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State the type of reaction, and the reagent(s) and conditions required.

(3)

Type

.....

Reagent(s)

.....

Conditions

.....

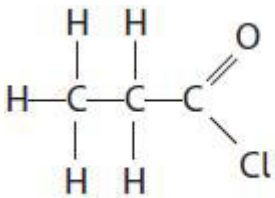
(Total for question = 9 marks)

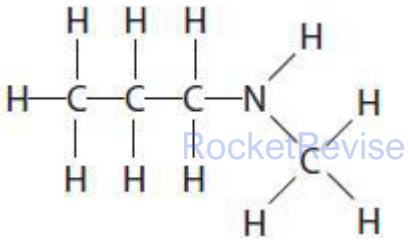
Q57.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Propanoyl chloride and *N*-methylpropylamine react to form an amide and hydrogen chloride.

The structures of the reactants are shown.



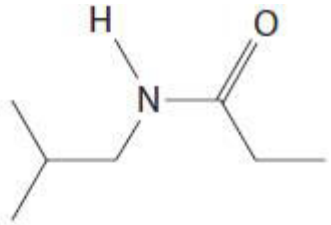


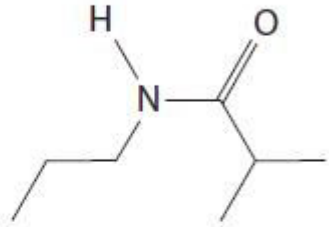
(a) What is the structure of the amide formed?

(1)

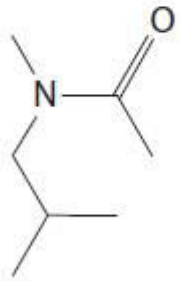
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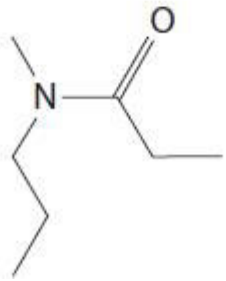
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☒ **A** 

☒ **B** 

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☒ **C** 

☒ **D** 

(b) If 64.3 g of the amide is produced, what volume of HCl is formed at room temperature and pressure (r.t.p.)?

Assume the stoichiometry is 1 : 1.
1 mole of gas occupies 24.0 dm^3 at r.t.p.

(1)

☒ **A** 12.0 dm^3
☒ **B** 12.2 dm^3
☒ **C** 23.9 dm^3
☒ **D** 120 dm^3

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(Total for question = 2 marks)

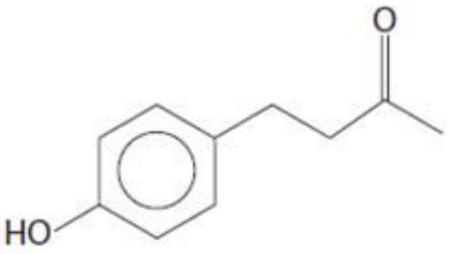
Q58.

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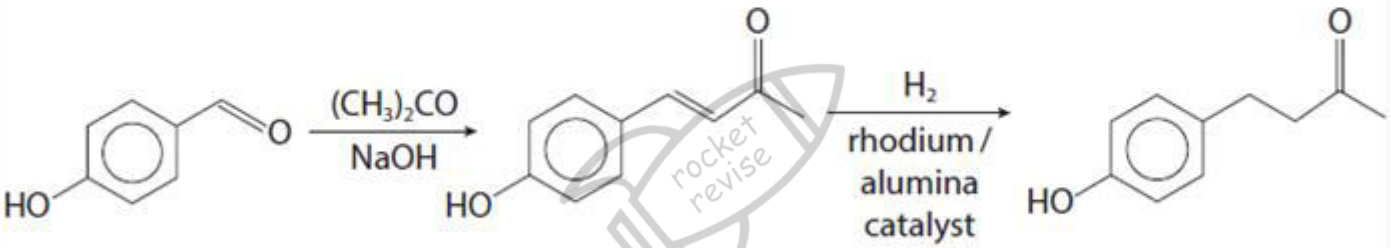
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Raspberry ketone

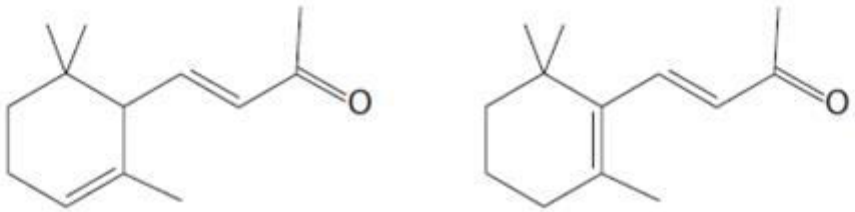
Over two hundred different compounds contribute to the smell of raspberries, including the one known as *raspberry ketone*.



Raspberry ketone is also found in other fruits such as cranberries and blackberries. It can be extracted from fruit but it is present in a very low percentage and so it is prepared industrially by a variety of methods.
The two-step method shown gives a 99% yield.



Two other compounds which contribute to the characteristic smell of raspberries are shown.

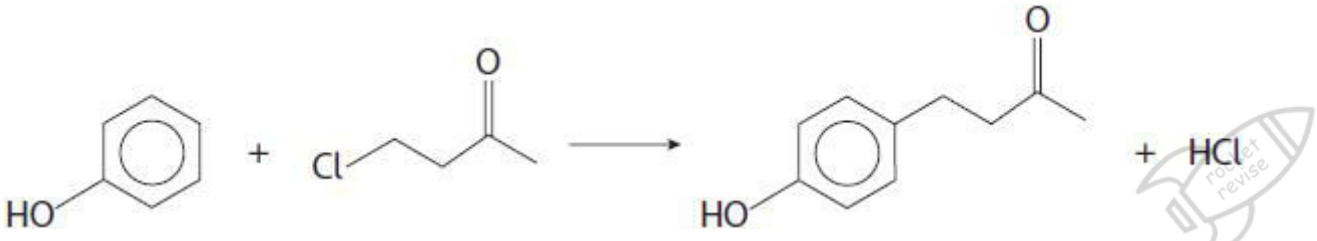


α-ionone β-ionone

(a) Give the **empirical** formula of the *raspberry ketone*.

(1)

(b) *Raspberry ketone* can be synthesised in the laboratory from phenol using a Friedel–Crafts reaction. The overall reaction is shown.



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(i) Draw the mechanism for this reaction, using appropriate curly arrows.
Include equations showing the formation of the electrophile and the regeneration of the catalyst at the end of the reaction.

(5)

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(ii) Give a reason why the yield of the *raspberry ketone* by this method is **not** high.

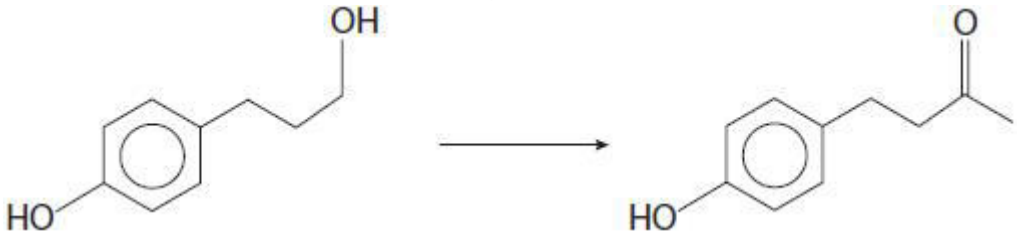
(1)

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(c) *Raspberry ketone* can also be synthesised in the laboratory using a Grignard reagent via a multi-step process. The overall reaction is shown.



Devise this multi-step synthesis.
Give the reagents and conditions for each step in the synthesis, including the formation of the Grignard reagent and the structures of the intermediates.

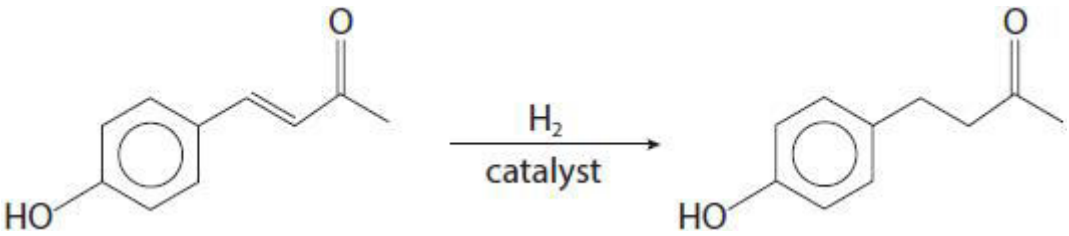
(7)

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(d) The second step of the industrial preparation of the *raspberry ketone* involves the reduction shown.

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Suggest why the choice of reducing agent is particularly important in this reaction.

(1)

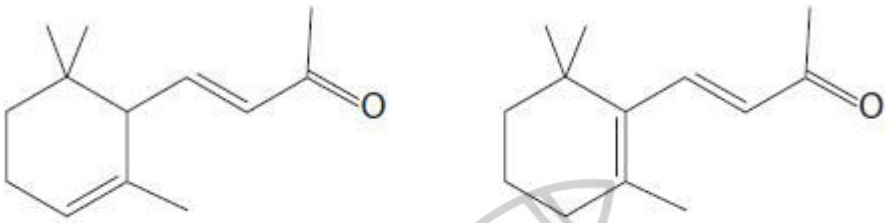
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(e) The α-ionone and β-ionone molecules are structural isomers.

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α-ionone β-ionone

(i) Compare and contrast the **stereoisomerism** of these two molecules.
On the structures, circle any features of the molecules that give rise to stereoisomerism.

(3)

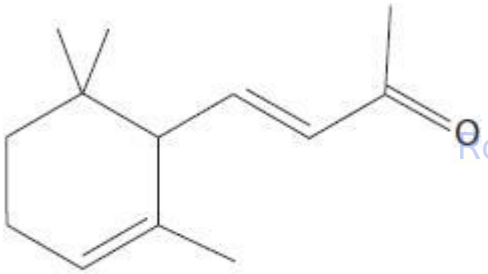
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(ii) Identify with an asterisk (*) the carbon atom environment in α-ionone which would be expected to produce a peak with a **quartet** splitting pattern in a high resolution proton NMR spectrum.

(1)



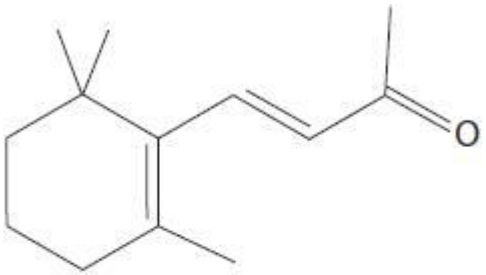
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α-ionone

(iii) Predict the number of peaks in the carbon-13 NMR spectrum of β-ionone, which has the molecular formula $C_{13}H_{20}O$.
Justify your answer.

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β-ionone

(1)

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.....

(Total for question = 20 marks)

Q59.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The formula of a chromium compound is $[Cr(H_2O)_4Cl_2]^+Cl^-$.

The molar mass of the **compound** is 230.5 g mol^{-1}.

What is the percentage by mass of chlorine in the **complex ion**?

- ☐ **A** 18.2%
- ☐ **B** 30.8%
- ☐ **C** 36.4%
- ☐ **D** 46.2%

(Total for question = 1 mark)

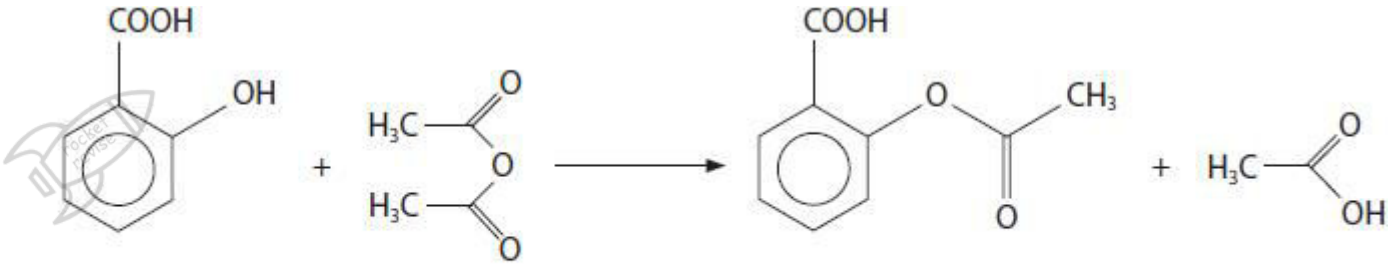
Q60.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

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A student prepared a sample of aspirin. The equation for the reaction is shown.

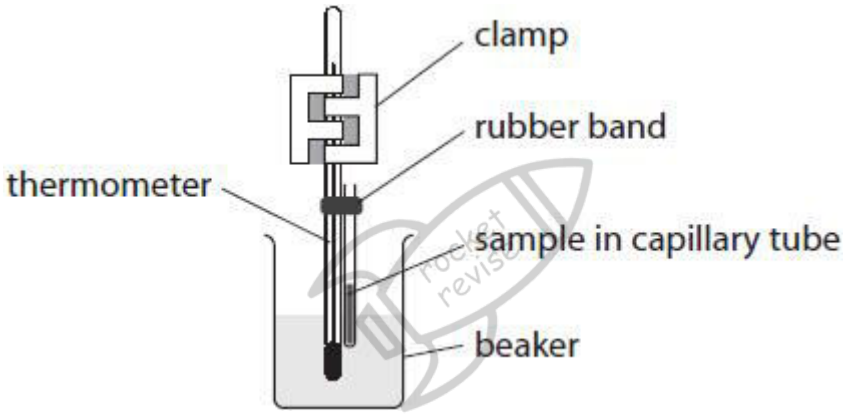


2-hydroxybenzoic acid + ethanoic anhydride → aspirin + ethanoic acid

The aspirin was recrystallised using an ethanol-water mixture as the solvent.

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The student determined its melting temperature using the apparatus shown.



The student's sample melted over a range of 8 degrees starting at 125 °C.

The published melting temperature of pure aspirin is 135 °C.

(a) Which is a suitable liquid to go in the beaker?

(1)

- ☒ **A** deionised water
- ☒ **B** ethanol
- ☒ **C** hexane
- ☒ **D** mineral oil

(b) Which is likely to be the main impurity in the sample of aspirin?

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(1)

- ☒ **A** ethanoic acid
- ☒ **B** ethanoic anhydride
- ☒ **C** ethanol
- ☒ **D** 2-hydroxybenzoic acid

(Total for question = 2 marks)

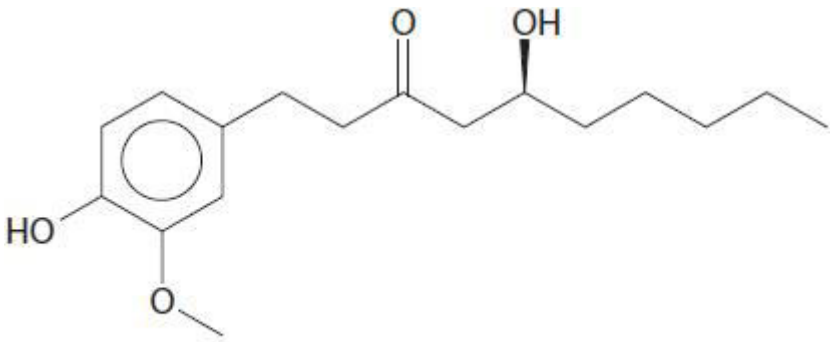
Q61.

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(a) Gingerol is a compound found in fresh ginger that activates spice receptors on the tongue, giving raw ginger a hot taste.

The skeletal formula of gingerol is shown.



gingerol

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(i) Give the molecular formula of gingerol.

(1)

.....

(ii) The OH group is shown attached to the carbon chain by a wedge-shaped bond.
Suggest why the bond between the carbon chain and the OH group is shown as a wedge.

(2)

.....

.....

.....

.....

.....

(b) Cooking fresh ginger converts gingerol into zingerone, which is less pungent and has a sweeter flavour.

Zingerone can be formed in a four-step synthesis from coniferyl alcohol. Step **2** in the synthesis involves a Grignard Reagent, while Steps **1**, **3** and **4** are redox reactions.
The synthesis is shown with the structures of the intermediate compounds incomplete.
Complete this four-step synthesis of zingerone from coniferyl alcohol.
Include in your answer completed structures of the intermediate compounds and the reagents and conditions required.

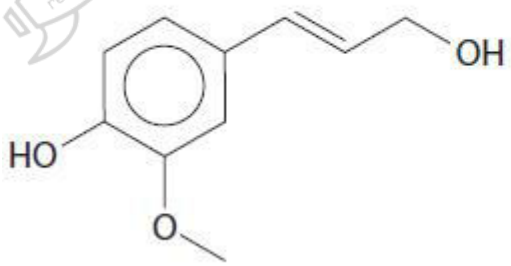
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(7)

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Step **1**

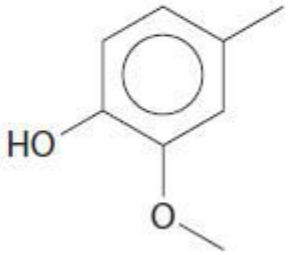


coniferyl alcohol

Reagents and conditions

.....

.....



intermediate 1

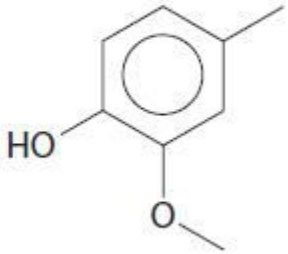
Step **2**

Reagents and conditions

.....

.....

intermediate 1



intermediate 2

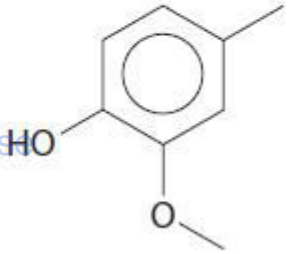
Step **3**

Reagents and conditions

.....

.....

intermediate 2

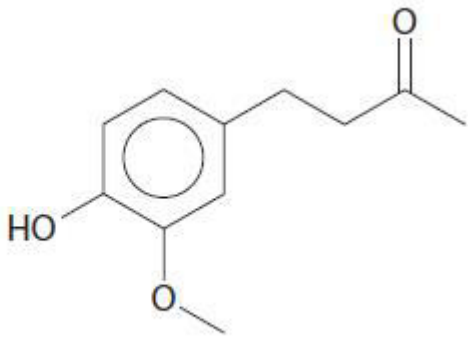


intermediate 3

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Step **4**



Reagents and conditions

.....

.....

intermediate 3 → zingerone

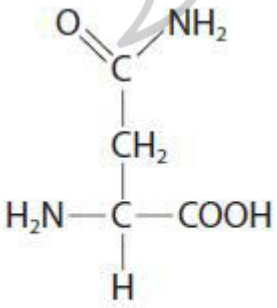
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(Total for question = 10 marks)

Q62.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Asparagine is an amino acid.



asparagine

(a) What will be the structure of asparagine in an aqueous solution at pH 2?

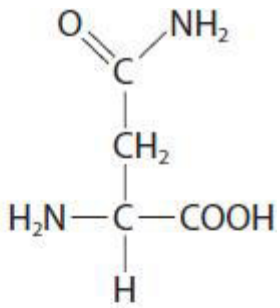
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(1)

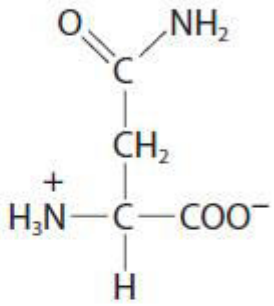
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☒ **A**

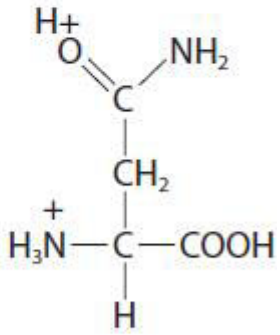


☒ **B**

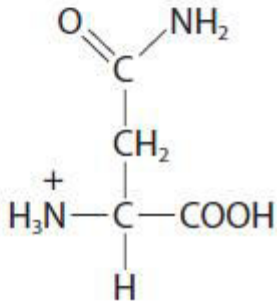


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☒ **C**

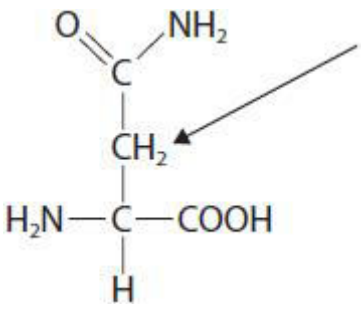


☒ **D**



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(b) In a high-resolution proton NMR spectrum of asparagine, what would be the splitting pattern for the protons indicated by the arrow?



(1)

☒ **A** singlet

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☒ **B** doublet
☒ **C** triplet
☒ **D** quartet

(Total for question = 2 marks)

Q63.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which sequence describes recrystallisation?

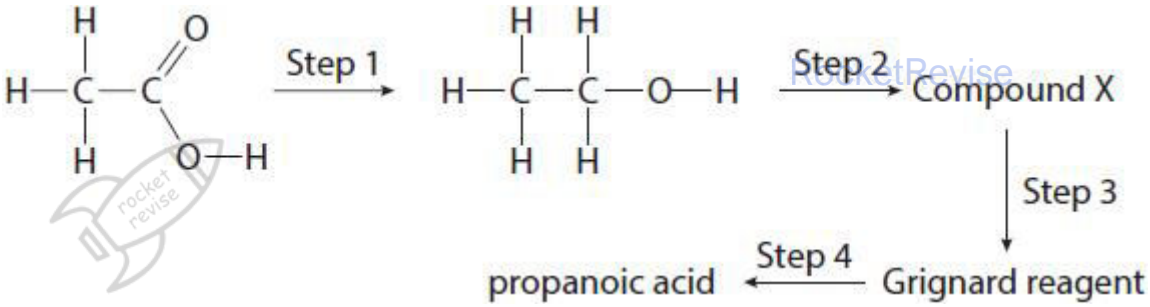
☒ **A** dissolve in the minimum volume of cold solvent, filter to remove the insoluble impurities, then add hot solvent and filter to remove the soluble impurities
☒ **B** dissolve in the minimum volume of hot solvent, filter to remove the soluble impurities, wash with cold solvent to remove the insoluble impurities
☒ **C** dissolve in the minimum volume of hot solvent, filter to remove the insoluble impurities, then cool and filter to remove the soluble impurities
☒ **D** dissolve in the minimum volume of hot solvent, filter to remove the soluble impurities, then cool and filter to remove the insoluble impurities

(Total for question = 1 mark)

Q64.

Grignard reagents are used to increase the length of the carbon chain in organic molecules.

An outline synthesis of propanoic acid from ethanoic acid is shown.



Complete the table.

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Step **1** reagent(s) and conditions	
Step **2** reagent(s)	
Name or formula of compound X	
Step **3** reagent(s) and conditions	
Formula of the Grignard reagent	
Step **4** reagent(s)	

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(Total for question = 6 marks)

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Q65.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An 8.4 g sample of a hydrocarbon formed 26.4 g of carbon dioxide on complete combustion. What is the molecular formula of this compound?

- ☐ **A** CH_2
- ☐ **B** C_6H_6
- ☐ **C** C_2H_6
- ☐ **D** C_6H_{12}

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(Total for question = 1 mark)

Q66.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What are the volumes of oxygen required **and** carbon dioxide produced, at room temperature and pressure (r.t.p.), on complete combustion of 5.8 g of butane, C_4H_{10}?

[Molar volume of a gas at r.t.p. = 24.0 dm^3 mol^{-1} M_r C_4H_{10} = 58.0]

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		volume of oxygen / dm^3	volume of carbon dioxide / dm^3
☐	**A**	15.6	9.6
☐	**B**	12.0	9.6
☐	**C**	21.6	2.4
☐	**D**	15.6	12.0

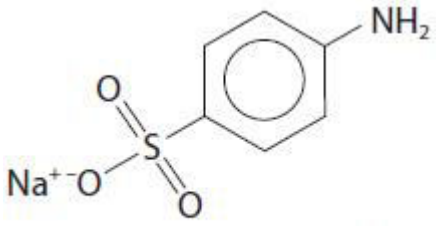
(Total for question = 1 mark)

Q67.

Sunset yellow is a common food colouring.

Sunset yellow can be produced in a coupling reaction starting with compound **A**.

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compound **A**

(a) State the reagents and conditions required for the formation of a diazonium ion from compound **A**.

(2)

.....

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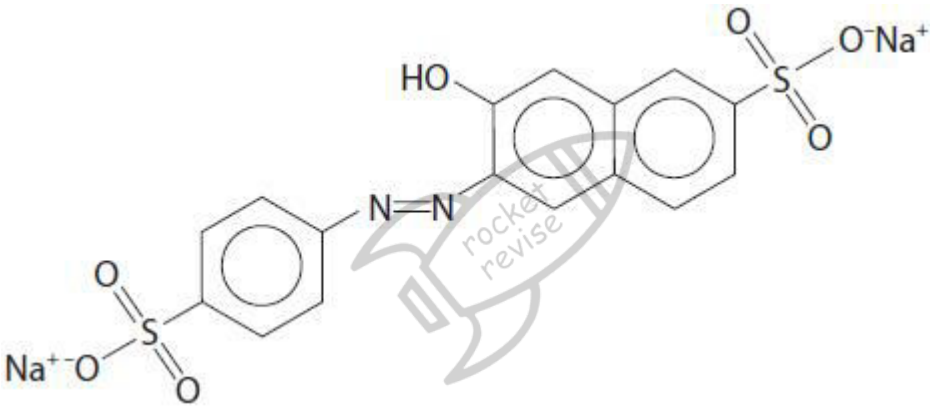
(b) Draw the structure of the diazonium ion formed.

(1)

(c) The structure of sunset yellow is shown.

Use a circle to identify the functional group, other than the diazonium group, that is essential for the coupling reaction to occur.

(1)



(d) (i) After sunset yellow is formed, it needs to be purified.

Name the process that is used.

(1)

.....

(ii) The yield, by mass, for the formation of sunset yellow is 67.9%.
Calculate the mass of sunset yellow produced from 5.00 g of compound **A**.
[M_r of sunset yellow = 452.2]

(3)

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(Total for question = 8 marks)

Q68.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

This question is about branched alkanes.

(a) How many peaks are there in the ^{13}C NMR spectrum of 2,2-dimethylpropane?

(1)

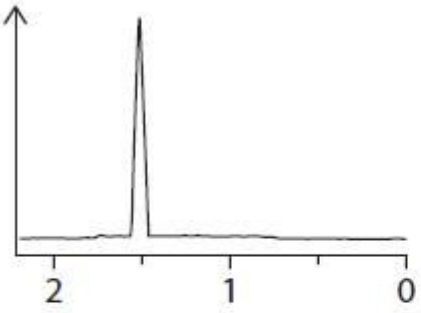
- ☐ **A** one
- ☐ **B** two
- ☐ **C** three
- ☐ **D** five

(b) Four students sketched the high-resolution proton NMR spectrum of 2,3-dimethylbutane.

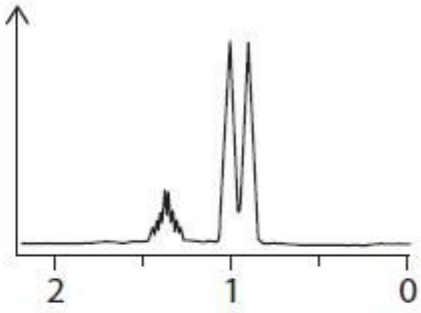
Which sketch is most likely to be correct?

(1)

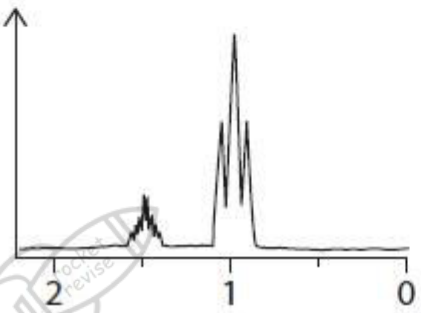
☐ **A**



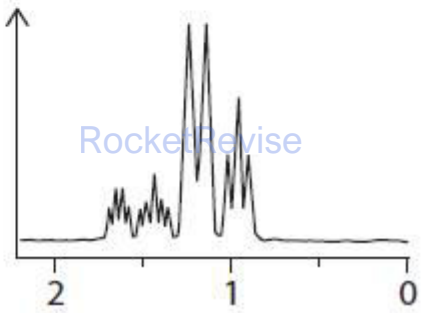
☐ **B**



☐ **C**



☐ **D**



(Total for question = 2 marks)

Q69.

Answer the question with a cross in the box you think is correct ☒. If you change your mind

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about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

A substituted benzene compound has the general formula C_nH_{2n-6}.

When completely combusted, 0.025 mol of the compound produced 11.0 g of CO_2.

Which could be the M_r of the compound?

- ☒ **A** 64
- ☒ **B** 78
- ☒ **C** 128
- ☒ **D** 134

(Total for question = 1 mark)

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